An Ex-ante Evaluation of the Economic Impact of the African Continental Free Trade Area (AfCFTA)*

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Abstract

Since the announcement that trading was to commence under the AfCFTA agreement on 1 January 2021, full trading has been delayed due to challenges in finalizing tariff concessions and Rules of Origin requirements. In anticipation of the full commencement of trading under the agreement, numerous studies have emerged, mostly focusing on the simulation of tariff-based scenarios. The focus of these studies only captures observable trade barriers, neglecting the substantial unobservable frictions that affect cross-border exchange in Africa. Motivated by this limitation, I examine the ex-ante impact of the AfCFTA by employing bilateral border indicators within a structural gravity framework to examine how the elimination of the intra-AfCFTA border wedge would influence trade and welfare. I find evidence of substantial but heterogeneous potential trade creation and welfare gains across sectors, industries, and countries. Specifically, the partial equilibrium estimates suggest that sectors and industries currently facing higher border frictions are likely to realize disproportionately larger trade gains, ranging from 17% for mining and energy to 42% for services at the sectoral level and from 6% up to 115% at the industry-level, following the full implementation of the AfCFTA agreement. The border-based general equilibrium welfare estimates demonstrate significant heterogeneity, with services showing the largest gains (up to 250%), followed by mining and energy (up to 158%), agriculture (up to 93%), and manufacturing experiencing the smallest gains (up to 57%). This border-based experiment is best interpreted as an upper-bound scenario that captures the long-run potential of continental integration under deep and effective implementation of the AfCFTA agreement. These findings provide a compelling economic case for expediting and prioritizing the full implementation of the AfCFTA despite current challenges, as doing so could substantially advance Africa's economic development and integration objectives outlined in Agenda 2063.

Keywords: AfCFTA agreement; Structural gravity; Unobservable trade costs; Partial equilibrium; General equilibrium.

JEL Classification Codes: F1; F10; F13; F14; F15.

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1 Introduction

At an Extraordinary Summit of the African Union (AU) Assembly on 5 December 2020, it was approved that trading under the African Continental Free Trade Area (AfCFTA) agreement would commence on 1 January 2021 (Tralac, 2025). Five years later, the objective of liberalizing intra-African trade has yet to be realized due to delays in finalizing tariff concessions and Rules of Origin (RoO). In response, the AfCFTA Secretariat launched the Guided Trade Initiative (GTI) on 7 October 2022 as a pilot mechanism to kick-start trading; however, participation remained limited to 11 African countries until its conclusion in April 2025 (Tralac, 2025).

Amid the full implementation delay, this article provides an ex-ante quantification of the potential trade and welfare effects of the AfCFTA agreement. Following recent developments in structural gravity literature (see Anderson et al., 2018; Larch et al., 2023), I make two significant contributions to the AfCFTA literature. First, I implement a partial equilibrium structural gravity model to quantify the ex-ante impact of the AfCFTA on intra-African trade. Specifically, I construct a bilateral border indicator that maps international trade between AfCFTA members within a two-way fixed effects structural gravity model that also includes standard gravity covariates and bilateral border indicators for AfCFTA exports to non-AfCFTA members and for exports between the rest of the world. The AfCFTA bilateral border indicator captures the residual trade frictions associated with international trade between AfCFTA members after accounting for standard gravity covariates, reflecting both observable and unobservable barriers to cross-border exchange. More importantly, the coefficient of this border indicator can be used to obtain not only the average existing border effects on intra-AfCFTA trade flows but also the potential trade creation resulting from the full implementation of the AfCFTA agreement.[1] Relying on this approach, I estimate the partial equilibrium structural gravity model at the sector- and industry-level covering 170 industries across four broad sectors, namely agriculture, manufacturing, mining & energy, and services. This granular approach offers policymakers a comprehensive understanding of the trade creation effects of the AfCFTA across sectors and industries, providing insights into where the greatest trade gains might materialize once RoO challenges are resolved.

Second, I quantify the potential sector-level welfare impacts of the AfCFTA agreement using a border-based general equilibrium counterfactual implemented through the General Equilibrium Poisson Pseudo–Maximum Likelihood (GEPPML) procedure proposed by Anderson et al. (2018). GEPPML provides a theoretically consistent and computationally tractable way to evaluate trade-policy counterfactuals in general equilibrium, while operationalizing the exact-hat structural gravity logic of Dekle et al. (2007, 2008). The analysis proceeds in three steps. First, I estimate a structural gravity model for a baseline year to recover the intra-African border wedge associated with AfCFTA member trade, as well as the exporter and importer fixed effects that implicitly capture baseline multilateral resistance terms. Second, I construct a counterfactual scenario that simulates

[1] For instance, in the PPML gravity framework, the coefficient on the AfCFTA border indicator, say β, represents a semi-elasticity such that the presence of an international border reduces trade by $(\exp(\beta) - 1) \times 100$ percent relative to domestic trade, while in the counterfactual analysis, the removal of this border wedge increases trade by $(\exp(-\beta) - 1) \times 100$ percent relative to the status quo.

full AfCFTA implementation by eliminating the intra-African international border wedge, while holding all other trade-cost determinants fixed. Conditional general equilibrium effects are then obtained by re-estimating the gravity equation with the counterfactual trade costs imposed as an offset, allowing multilateral resistance terms to adjust while nominal output and expenditure remain fixed. Third, I compute full endowment general equilibrium effects by allowing output and expenditure to respond endogenously to the border-removal shock through an iterative GEPPML procedure. Welfare effects are measured as changes in real income, capturing both the direct gains from reduced trade frictions and the indirect general equilibrium adjustments transmitted through prices, trade flows, and multilateral resistance terms. This general equilibrium analysis is particularly relevant in the current implementation phase of the AfCFTA, as it quantifies the long-run welfare potential of fully eliminating intra-AfCFTA border frictions and provides an economic benchmark for sustaining policy commitment during ongoing negotiations on RoO.

This border-based approach follows the recent ex-ante quantification method prescribed by Larch et al. (2023). Specifically, I adopt this approach because it helps capture the combined effect of observable and unobservable trade frictions, including tariffs, regulatory barriers, trade facilitation constraints, and service-related costs, that differentiate international from domestic trade. This approach enables an ex-ante evaluation of the AfCFTA that does not rely on often incomplete tariff and non-tariff policy measures, which capture only a limited subset of total trade costs and are subject to substantial measurement error (see Anderson and Van Wincoop, 2004; Borchert et al., 2014; Looi Kee et al., 2009, for studies on the limitations of tariffs and their measurement error). Moreover, it provides a unified framework that applies consistently to both goods and services trade, where conventional tariff data are particularly sparse. The analysis corresponds to the elimination of the intra-AfCFTA border wedge among AfCFTA members. While AfCFTA's legal commitments focus on partial tariff liberalization, the border-based experiment is best interpreted as an upper-bound scenario that captures the long-run potential of continental integration under deep and effective implementation. This perspective is particularly relevant given persistent implementation delays and the prominence of non-tariff and behind-the-border frictions in African trade. By focusing on the full border wedge, the analysis isolates the maximum scope for trade creation and welfare gains that could be achieved if AfCFTA succeeds in substantially reducing both formal and informal barriers to intra-African exchange.

The partial equilibrium estimates reveal substantial border effects and associated trade creation impacts of the AfCFTA on intra-AfCFTA trade at both the sectoral and industry levels. Sectoral evidence indicates that the presence of international borders reduces intra-AfCFTA trade by approximately 30% in services, 26% in agriculture, 16% in manufacturing, and 15% in mining and energy. This pattern suggests that intra-AfCFTA trade in services experiences the largest border effects, followed by agriculture, while border effects are more moderate in the manufacturing and mining sectors. The counterfactual analysis further shows that sectors exhibiting larger border effects also experience greater potential trade creation following border removal. Specifically, the elimination of borders between AfCFTA members is estimated to increase intra-African trade by about 42% in services, 34% in agriculture, 19% in manufacturing, and 17% in mining and energy. Evidently, these results indicate that AfCFTA implementation will have differentiated impacts across sectors, with services and agriculture experiencing the largest trade expansion due to their currently high border frictions, while manufacturing and mining are expected to see more modest gains.

The industry-level estimates reveal substantial and highly heterogeneous border effects, ranging from approximately 6% to 54% across industries. The largest border-reducing trade effects are observed in Raw and refined sugar and sugar crops (54%), Jewellery and related articles (43%), Eggs (40%), Dressing and dyeing of fur; processing of fur (39%), and Telecommunications, computer, and information services (39%), spanning the agriculture, manufacturing, and services sectors. In contrast, the smallest border effects are recorded in Basic chemicals except fertilizers (10%), Machinery for mining and construction (10%), Medical, surgical, and orthopaedic equipment (9%), Man-made fibres (6%), and Other articles of paper and paperboard (6%), all of which belong to the manufacturing sector. The counterfactual analysis indicates that eliminating borders between AfCFTA members would translate these large border wedges into substantial trade creation. Specifically, trade in the top five industries is projected to increase by approximately 115% for Raw and refined sugar and sugar crops, 75% for Jewellery and related articles, 68% for Eggs, 65% for Dressing and dyeing of fur; processing of fur, and 63% for Telecommunications, computer, and information services. By contrast, the bottom five industries experience more modest gains, ranging from 6% to 12%. These results indicate that industries currently facing higher border frictions stand to realize disproportionately larger trade gains following the full implementation of the AfCFTA agreement.

Finally, the welfare estimates suggest large gains from general equilibrium estimation. Specifically, they predict substantial heterogeneity across sectors and countries, with the services sector exhibiting the most pronounced gains (reaching up to 250% for Benin and down to 7% for Nigeria), followed by mining and energy (up to 158% for Benin and down to 5% for Angola), agriculture (up to 93% for Botswana and down to 13% for Egypt), and manufacturing (up to 57% for Lesotho and down to 12% for South Africa). These estimates exhibit a consistent pattern across all sectors, where smaller economies generally experience larger welfare gains than their larger counterparts, which aligns with the well-established inverse relationship between country size and welfare gains in general equilibrium literature. Accordingly, these welfare estimates should be interpreted as the long-run potential gains from deep continental integration under effective AfCFTA implementation, rather than as short-run effects of tariff liberalization alone. As a result, welfare gains arise not only from lower prices due to tariff removal but also from reductions in regulatory, informational, logistical, and services-related frictions that reallocate production and expenditure across sectors and trading partners.

This research relates to the growing literature on ex-ante evaluation of the AfCFTA agreement. This literature has evolved along two main strands, each with distinct methodological approaches and limitations. The first strand employs computable general equilibrium (CGE) and partial equilibrium (PE) models to simulate the effects of merchandise tariff elimination, often complemented with partial reductions in non-tariff barriers (Mevel and Karingi, 2012; Jensen and Sandrey, 2015; Chauvin et al., 2017; Abrego et al., 2019; Simola et al., 2021; Bayale et al., 2022; Fusacchia et al., 2022; Oyelami, 2022). Early studies by Mevel and Karingi (2012) and Chauvin et al. (2017) using MIRAGE models found that combining tariff and NTB liberalization scenarios yields substantially larger gains than tariff-only scenarios, with welfare gains ranging from 0.20% to 2.64% and intra-African trade increases reaching up to 140% for some countries. More recently, Abrego et al. (2019) project welfare gains of 2% to 4% and trade flow increases of 78% to 82%, while sector-specific analyses by Simola et al. (2021) and Fusacchia et al. (2022) focus on agri-food industries. However, a key limitation of this approach is that merchandise tariffs represent only a small fraction of overall trade restrictions between AfCFTA members compared to NTBs and trade facilitation challenges (MacLeod et al., 2023). As Larch et al. (2023) argue, even when observable NTB measures are simulated, the estimates still account for the lower bound of potential gains from comprehensive trade liberalization.

The second strand adopts a different strategy by using ex-post estimates of existing Regional Economic Communities (RECs) in Africa as policy proxies within structural gravity models (Geda and Yimer, 2023; Fofack et al., 2021; Stack et al., 2024). Geda and Yimer (2023) incorporate dummies for four major African RECs in a PPML gravity model and find net trade creation ranging from 42.6% to 60.75% increase in intra-African exports. Stack et al. (2024) use an AfCFTA indicator variable and estimate a 77% increase in intra-African manufacturing exports, while Fofack et al. (2021) project trade increases of 24% to 25% and welfare gains of 0.12% to 0.19% using a dynamic GE-PPML framework. While this approach aligns with current best practices for trade policy evaluation, it is more suitable for ex-post rather than ex-ante analysis. As Larch et al. (2023) note, estimating policy effects based on external samples and historical relationships from existing RTAs fails to capture the unique structural features and policy innovations embedded in new agreements.

I contribute to this literature by adopting the bilateral border variable approach within the structural gravity framework, following Larch et al. (2023), to capture both observable and unobservable trade frictions that would be eliminated under the AfCFTA. This methodology offers several advantages over existing approaches: (i) it does not require patchy tariff or NTB data, (ii) captures all trade frictions not observed by standard gravity covariates, (iii) accounts for trade costs in the services sector where tariff data is unavailable, and (iv) can be implemented for both partial and general equilibrium estimation. My analysis extends existing work in two key dimensions. First, I provide industry-level granularity covering 170 industries across four broad sectors, expanding beyond the limited agri-food focus of Simola et al. (2021) and Fusacchia et al. (2022) and the sector-level analyses of Mevel and Karingi (2012), Chauvin et al. (2017), and Stack et al. (2024). Second, I quantify border-based general equilibrium welfare effects that account for both observable and unobservable trade costs across all sectors, offering more comprehensive projections than studies that focus primarily on tariff liberalization scenarios (Abrego et al., 2019; Simola et al., 2021).

The remainder of this paper proceeds as follows: Section 2 provides a brief overview of the AfCFTA agreement. Section 3 presents the data, descriptive analysis, and empirical strategy. Section 4 discusses the interpretation and discussion of the partial and general equilibrium analyses. Section 5 presents the robustness checks of the main findings, and Section 6 concludes with the policy relevance.

2 A Brief Overview of the AfCFTA Agreement

The AfCFTA is a continent-wide trade agreement that is part of the broader blueprint and master plan of the AU's "Agenda 2063: The Africa We Want"—a pan-African vision to transform the continent into the world's future powerhouse (AfCFTA, 2024). The AfCFTA is the world's largest free trade area (FTA) since the World Trade Organization (WTO) was established, unifying all 55 African States and the existing eight RECs (AfCFTA, 2024).[2] Its main objective is to create a single, liberalized market that not only facilitates the free mobility of goods and services but also capital and labor to effectively deepen African economic integration (Tralac, 2025). Specifically, the aim is to progressively liberalize 97% of tariffs on intra-African trade in goods categorized as non-sensitive products, of which 7% are classified as sensitive products to be liberalized over a longer timeframe. Countries are allowed to exclude the remaining 3% of tariffs from liberalization to protect specific national concerns. Overall, the period of tariff liberalization will take five years for non-least developed nations and ten years for least developed countries. According to MacLeod et al. (2023), the AfCFTA can be classified as a deep trade agreement because it extends beyond the elimination of tariffs, covering non-tariff barriers, provisions on trade facilitation, trade in services, investment, intellectual property rights, competition policy, digital trade, and recognition of women and youth in trade.

The AfCFTA emerged from a series of negotiations that began in June 2015 to establish a continental single market (see Figure 1 for the full timeline of the AfCFTA milestones). Its consolidated pact was signed by 44 out of the 55 African Union (AU) States during an extraordinary summit held in Kigali, Rwanda, between 17 and 21 March 2018, with Eritrea being the only African State that has not signed the agreement (Tralac, 2025). The agreement entered into force on 30 May 2019 for the 24 signatories that submitted their instruments of ratification, and this number increased to 48 out of 54 signatories as of August 2024.[3] Although the AU Assembly approved that the commencement of trading under the AfCFTA should start on 1 January 2021, full trading has yet to materialize as of December 2025 (Tralac, 2025).

The primary obstacle to full trading under the agreement is the delay in finalizing negotiations on tariffs and RoO (Tralac, 2025). RoO are the criteria used to determine the national source of a product, establishing which goods qualify for preferential treatment under trade agreements by specifying the minimum local processing or content requirements. According to Tralac (2025), about 92.3% of tariff lines have already been agreed. To kick-start trading among AfCFTA States, the AfCFTA Secretariat launched the Guided Trade Initiative (GTI) on 7 October 2022 as a pilot phase and a solution-based approach through which the AfCFTA agreement will become fully op-

[2]The eight RECs in Africa are the Arab Maghreb Union (AMU), the Common Market for Eastern and Southern Africa (COMESA), the Community of Sahel-Saharan States (CEN-SAD), the East African Community (EAC), the Economic Community of Central African States (ECCAS), the Economic Community of West African States (ECOWAS), the Intergovernmental Authority on Development (IGAD), and the Southern African Development Community (SADC). These eight RECs, which range from customs unions to FTAs, serve as the building blocks for the institutional framework of the AfCFTA to ensure the coordination of measures for resolving non-tariff barriers (NTBs), harmonizing standards, and monitoring implementation (UNECA, 2019).

[3]See Table C.1 for the full list of AfCFTA States and their ratification status.

erational (Tralac, 2025). The initiative serves as an interim solution to enable trading for States that have met the minimum requirements for preferential trade under the agreement, to assess the preparedness of the private sector, and to test the efficiency of the operational, institutional, legal, and trade policy environment under the AfCFTA (Tralac, 2025). States Parties are permitted to join the GTI under the condition that their Provision Schedules of Tariff Concessions are verified and the RoO for the intended products are finalized (Tralac, 2025). Thirty-one countries that met these conditions joined the initiative, though only thirteen of them finalized the required legal modalities to enable trading under the AfCFTA as of January 2024, and as of September 2025, only eleven State Parties, namely Algeria, Cameroon, Ghana, Egypt, Kenya, Mauritius, Nigeria, Rwanda, South Africa, Tanzania, and Tunisia, are actively trading under AfCFTA rules (Tralac, 2025). The conclusion of the GTI was confirmed at the 16th Council of Ministers Meeting in April 2025, allowing for trade under AfCFTA rules for those countries successfully trading under the GTI to commence on a permanent basis (Tralac, 2025). The GTI has met its objectives and affirmed the effective implementation of the application of AfCFTA legal instruments for trade in goods among participating countries.

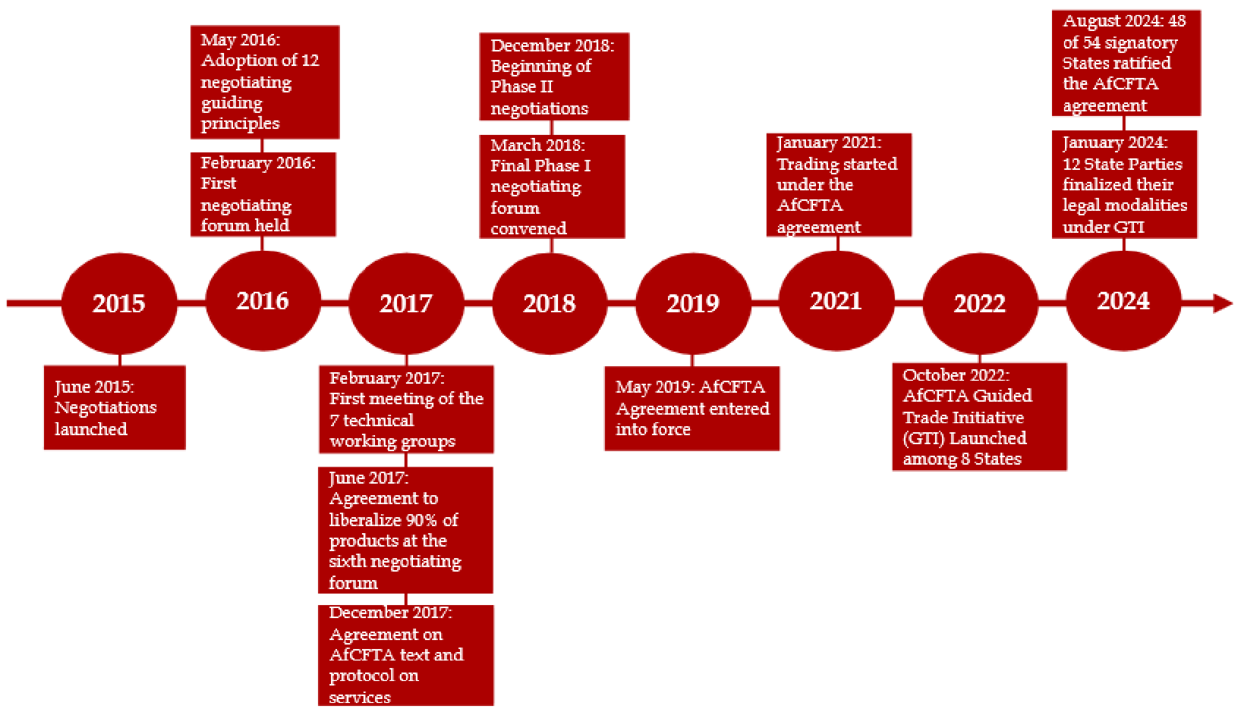


Figure 1: AfCFTA milestone timeline

Source: Author's construction based on the partly developed timeline by UNECA (2019) and information from Tralac (2025).

3 Methodology

This section presents the empirical strategy used to identify the AfCFTA border wedge and to quantify its trade and welfare implications within a structural gravity framework. The analysis combines a partial equilibrium estimation of bilateral border effects with a general equilibrium counterfactual implemented through GEPPML. The partial equilibrium analysis isolates the average intra-AfCFTA border wedge and computes the implied counterfactual trade creation from its removal. The general equilibrium analysis then embeds this border-removal shock into a structural gravity system, allowing multilateral resistance terms, prices, output, and expenditure to adjust endogenously, thereby yielding welfare effects consistent with exact-hat general equilibrium logic.

3.1 Data

This study employs consecutive year panel data for the ex-ante evaluation of AfCFTA within the structural gravity model. For the empirical analysis, I use data on bilateral trade flows and gravity covariates, while I source Effectively Applied (AHS) tariffs for descriptive analysis of trends in trade flows and barriers.

I employ bilateral trade flows from the International Trade and Production Database for Estimation Release 2 (ITPD-E-R02) by Borchert et al. (2021, 2022b). This database is well-suited to implement the border-based method due to its rich country-pair coverage, sectoral and industry-level disaggregation, and inclusion of all AfCFTA signatories for both intra- and international trade flows. The inclusion of domestic trade flows is particularly important for identifying border effects in structural gravity models. The database provides bilateral trade flows for 265 countries and 170 industries spanning the period from 1986 to 2019 classified across four broad sectors, namely *Agriculture* (1986–2019) covering 28 industries, *Manufacturing* (1988–2019) with 118 industries, *Mining & Energy* (1988–2019) with 7 industries, and *Services* (2000–2019) comprising 17 industries.

Given that the data for the Services sector spans from 2000 to 2019, I restrict the observation of the dataset for the broad sector and industry-level analysis to this period. For the PE analysis, I aggregate the data for each broad sector by summing the data for all the industries within the corresponding main sector, and I utilize all the country-pair data points in the database for the sector- and industry-level analysis. For computational reasons, I restrict the number of countries to 160 (all listed in Table D.2) for the GE analysis using the cross-sectional observation of the year 2019.[4] It is also worth noting that despite the comprehensiveness of the ITPD-E-R02 database, the dataset is highly unbalanced with a significant amount of missing observations for intranational trade flows. To implement the GE analysis, I transform the data into a balanced dataset and impute missing observations for intra-national trade following the three steps used by Larch et al. (2023). First, I construct the ratio of total exports to internal sales for each of the available industries. Second, I combine the ratios with the data on total exports of each country and the disaggregated industry to fill the missing domestic trade. Third, I aggregate the balanced observations to the four broad sectors, which I utilize in the GE analysis.

I also gather data on standard gravity covariates, namely distance, contiguity, colonial ties, common language, and RTA dummy, from the dynamic gravity dataset of the U.S. International

[4] I confirm the robustness of the general equilibrium results using an alternative database based on manufacturing sector data from the Structural Gravity Database by Larch et al. (2019).

Trade Commission (USITC) (Gurevich and Herman, 2018). The dataset provides annual data from 1948 to 2019 for 286 countries, which I match with the ITPD-E-R02 database. Then, I source effectively applied (AHS) tariff rates from the World Integrated Trade Solution (WITS) at the HS 6-digit product level for intra-African imports (tariffs imposed by African importers on goods from African exporters) from 2000 to 2019. I utilize the AHS tariffs because some African country pairs trade under preferential schemes. To align tariffs with the sectoral dataset, HS 6–digit products are mapped to the *Agriculture*, *Manufacturing*, and *Mining & Energy* aggregates using standard HS–to–industry concordances consistent with the ITPD sectoring. Sectoral tariff rates are constructed as trade-weighted averages at the reporter–partner–sector–year level.

3.2 Descriptive Analysis

Table 1 presents summary statistics for intra-African trade flows and gravity covariates. The dataset comprises 109,705 country-pair observations across four broad sectors. Panel A shows that manufacturing accounts for the largest share of observations (50,686) with a mean trade value of 17.31 million USD, followed by agriculture (34,389 observations, mean of 2.09 million USD), mining and energy (24,516 observations, mean of 9.32 million USD), and services (114 observations, mean of 36.36 million USD). The low median values close to zero across all sectors, combined with high standard deviations, indicate substantial heterogeneity in bilateral trade flows, with most country pairs trading little while a few exhibit exceptionally large trade values. Panel B reports gravity covariates. The average distance between African country pairs is 3,423 kilometers (median: 3,232 km), ranging from 158 to 9,760 kilometers. Among the binary variables, contiguity characterizes only 10% of country pairs, while colonial ties are present in just 1% of observations. In contrast, 56% of country pairs share a common official language, and 11% belong to the same regional trade agreement. These statistics reflect the linguistic legacy of colonialism and the fragmented nature of regional integration in Africa, where multiple overlapping trade agreements coexist.

Table 1: Summary Statistics of Intra-African Trade and Gravity Covariates

Variable/Sector	Obs.	Mean	Median	SD	Min	Max
Panel A: Trade by Sector (million USD)						
Agriculture	34389	2.09	0.00	14.05	0.00	699.26
Manufacturing	50686	17.31	0.03	136.08	0.00	5266.78
Mining and Energy	24516	9.32	0.00	111.20	0.00	5111.88
Services	114	36.36	0.00	82.92	0.00	452.75
Panel B: Gravity Covariates						
Distance	109705	3422.61	3232.34	1910.97	158.20	9760.23
Contiguity	109705	0.10	0.00	0.31	0.00	1.00
Colony	109705	0.01	0.00	0.08	0.00	1.00
Common Language	109705	0.56	1.00	0.50	0.00	1.00
RTA	109705	0.11	0.00	0.31	0.00	1.00

Note: Statistics are based on the author's computations. This table reports intra-African trade by sector based on ITPD-E-R02 data and gravity covariates based on the Dynamic Gravity Dataset.

Figure 2 displays the evolution of intra-African export (% of GDP) and AHS weighted average tariffs across four broad sectors from 2000 to 2019. Panel (a) shows that agricultural exports increased from around 5% of GDP in 2000 to close to 8% in 2017, while tariffs declined from above 15% to below 10%. Panel (b) illustrates that manufacturing trade grew from around 35% in 2000 to almost 90% of GDP in 2017, with tariffs decreasing from around 11% to 7%. Panel (c) depicts mining and energy trade exhibiting high volatility, ranging from near 5% in 2000 to over 20% of GDP in 2013 and dropping to 12% in 2017, with consistently low tariffs from around 6% in 2000 to roughly 3% in 2019. Panel (d) presents services trade showing substantial variation between 0% and 2% of GDP over a limited time span. Across all goods sectors, the general downward trend in tariffs has not been accompanied by commensurate increases in trade intensity, suggesting that non-tariff barriers and structural factors significantly constrain intra-African exports beyond conventional tariff measures. This highlights the importance of a comprehensive ex-ante evaluation that captures both observable and unobservable trade costs.

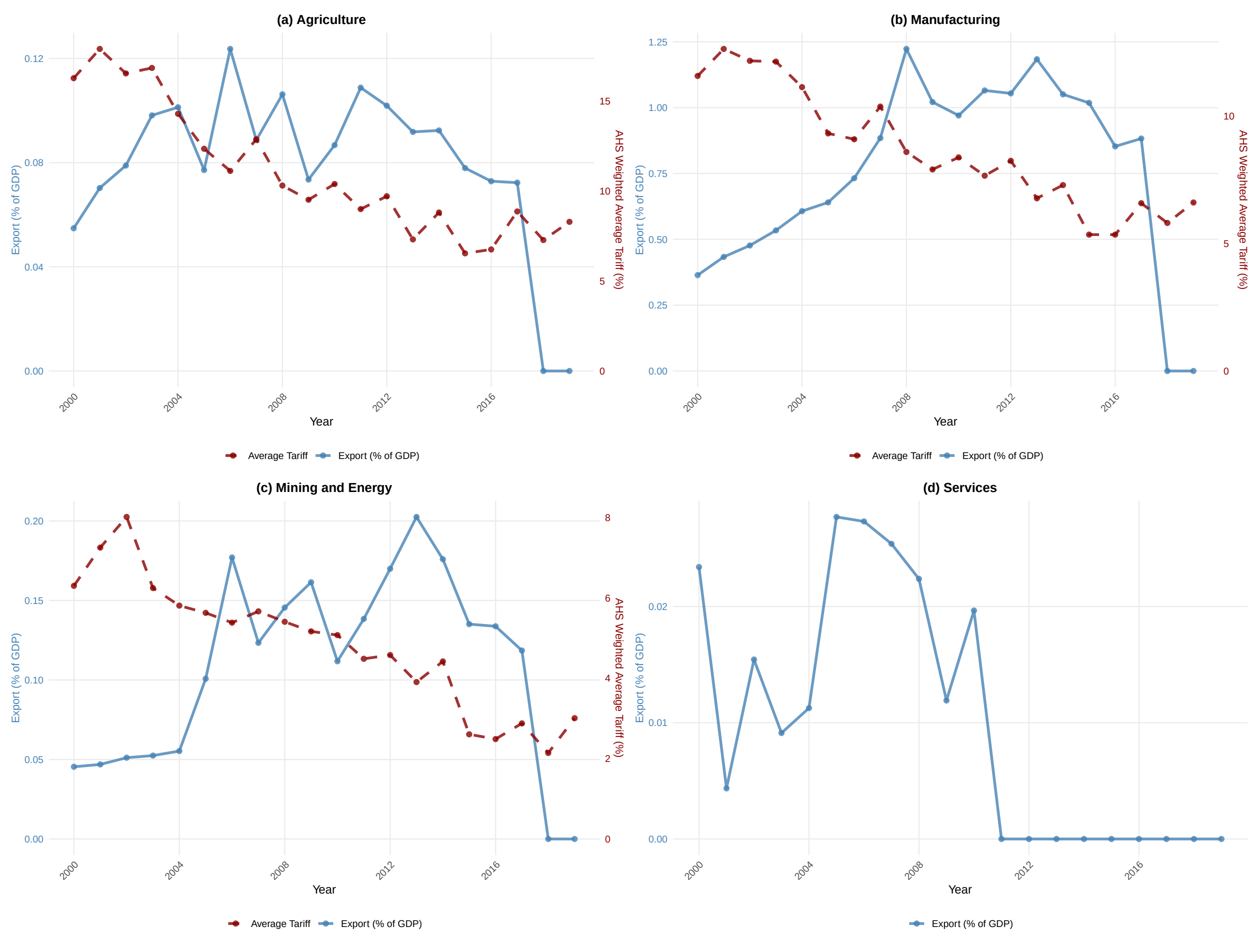


Figure 2: Intra-African Export (% of GDP) and Tariffs by Sector

Source: Author's computation based on export flows and exporter GDP data from ITPD-E-R02 and the tariff data retrieved from the WITS. This figure displays the evolution of intra-African exports (% of GDP) and AHS weighted average tariff (%) for Agriculture in panel (a), Manufacturing in panel (b), and Mining and Energy in panel (c), and for intra-African exports (% of GDP) for Services in panel (d).

3.3 Empirical Strategy

3.3.1 Partial Equilibrium Analysis

The structural gravity model has become the workhorse framework for quantifying the effects of trade policies and other determinants of bilateral trade flows, largely due to its strong predictive performance and rigorous microeconomic foundations (Larch et al., 2021; Yotov, 2024). This framework builds on the seminal contributions of Anderson and Van Wincoop (2003), who derive a gravity model of trade in goods differentiated by origin. The model is characterized by the following system of equations:

$$X_{ij,t} = \frac{Y_{i,t}E_{j,t}}{Y_t}\left(\frac{t_{ij,t}}{\Pi_{i,t}P_{j,t}}\right)^{1-\sigma}, \tag{1}$$

$$\Pi_{i,t}^{1-\sigma} = \sum_j \left(\frac{t_{ij,t}}{P_{j,t}}\right)^{1-\sigma}\frac{E_{j,t}}{Y_t}, \tag{2}$$

$$P_{j,t}^{1-\sigma} = \sum_i \left(\frac{t_{ij,t}}{\Pi_{i,t}}\right)^{1-\sigma}\frac{Y_{i,t}}{Y_t}. \tag{3}$$

where $X_{ij,t}$ denotes exports (intra $i = j$ and international $i \neq j$) from exporter i to importer j at time t; $Y_{i,t}$ and $E_{j,t}$ are total output of i and total expenditure of j, respectively; Y_t is the value of world output; $t_{ij,t}$ represents bilateral trade costs between i and j at time t; and σ is the elasticity of substitution across goods. $\Pi_{i,t}$ and $P_{j,t}$ are the multilateral resistance terms introduced by Anderson and Van Wincoop (2003), which capture the average trade costs faced by exporters and importers, respectively.

Following Larch et al. (2023), I translate equations (1)–(3) into an empirical specification to ex-ante quantify the economic impacts of the AfCFTA agreement at the sector and industry level, while incorporating recent developments in the structural gravity literature. The baseline estimating equation is given by:

$$X^u_{ij,t} = \exp\left[\pi^u_{i,t} + \chi^u_{j,t} + \eta^u \times \textbf{GRAVITY}_{ij} + \beta^u_1 RTA_{ij,t} + \beta^u_2 BRDR_{ij}\right] \times \epsilon^u_{ij,t}. \tag{4}$$

where $X^u_{ij,t}$ represents nominal export flows from exporter i to importer j for a given unit of analysis u, either at the sector (s) or industry (k) level, in year t. Consistent with theoretical gravity models, the estimation includes both domestic and international trade flows (see Yotov, 2012; Bergstrand et al., 2015; Heid et al., 2021; Yotov, 2022) over consecutive years (see Egger et al., 2022). The inclusion of domestic trade flows is essential for identifying border effects, as it allows the estimation of trade frictions associated with crossing national borders relative to intra-national trade. Equation (4) is estimated using the Poisson pseudo-maximum likelihood (PPML) estimator. PPML is employed following Santos Silva and Tenreyro (2006, 2021) because it naturally accommodates zero trade flows and is robust to heteroskedasticity in trade data. The terms $\pi^u_{i,t}$ and $\chi^u_{j,t}$ denote exporter–sector(industry)–time and importer–sector(industry)–time fixed effects, respectively, which absorb all time-varying country-specific factors affecting trade. These fixed effects control for multilateral resistance terms as implied by Anderson and Van Wincoop (2003), ensuring consistency with the structural gravity framework. While gravity models are sometimes estimated using three-way fixed effects (exporter-time, importer-time, and country-pair effects), pair fixed effects are not included here. This choice is deliberate, as the inclusion of border indicators allows bilateral trade frictions—both observable and unobservable—to be estimated explicitly rather than being absorbed by pair fixed effects.

The vector $\textbf{GRAVITY}_{ij}$ includes standard bilateral gravity covariates: log bilateral distance ($DIST_{ij}$) measures the geographical distance between exporter i and importer j; common official language ($LANG_{ij}$) represents a binary indicator that is 1 if countries i and j share a common official language; colonial ties (COL_{ij}) denotes a binary indicator that equals 1 if countries i and j have had a colonial relationship; and contiguity ($CNTG_{ij}$) represents a binary indicator that equals 1 if countries i and j share a common border. Although these variables are measured at the country-pair level, their effects may differ across sectors and industries; therefore, their coefficients (η^u) are allowed to vary by unit of analysis. In addition, $RTA_{ij,t}$ is a time-varying indicator for regional trade agreements. The coefficient β_1^u captures the average trade effect of RTAs, controlling for the existing network of trade agreements so that AfCFTA-specific effects are not confounded by overlapping preferential arrangements. The variable $BRDR_{ij}$ is an international border indicator that equals one for international trade flows and zero for intra-national trade. Its coefficient β_2^u captures the average border-related trade cost wedge after controlling for exporter-year and importer-year fixed effects. The error term $\epsilon_{ij,t}^u$ captures remaining variation in bilateral trade flows not explained by the model.

The primary objective of the analysis is to ex-ante quantify how the unification of African economies under the AfCFTA agreement affects intra-African trade flows. To isolate AfCFTA-specific border frictions, I augment the baseline specification with a set of mutually exclusive border indicators that distinguish trade among AfCFTA members from other international trade relationships. The augmented specification is:

$$\begin{aligned} X_{ij,t}^u = \exp\left[\pi_{i,t}^u + \chi_{j,t}^u + \eta^u \times \textbf{GRAVITY}_{ij} + \beta_1^u RTA_{ij,t} + \beta_2^u BRDR_{INTL_{ij}}\right] \times \\ \exp\left[\beta_3^u BRDR_{AfCFTA_{ij}} + \beta_4^u BRDR_{AfCFTA\rightarrow ROW_{ij}}\right] \times \epsilon_{ij,t}^u. \end{aligned} \tag{5}$$

where $BRDR_{AfCFTA_{ij}}$ is a bilateral border indicator equal to one for international trade between AfCFTA member countries and zero otherwise. Its coefficient β_3^u captures the average intra-African border effect over the sample period, reflecting observable and unobservable trade frictions that differentiate international from domestic trade among AfCFTA members.[5] The variable $BRDR_{AfCFTA\rightarrow ROW_{ij}}$ equals one for international trade flows from AfCFTA members to non-member countries and zero otherwise. Its coefficient β_4^u captures the average border effects faced by AfCFTA members when exporting to the rest of the world (ROW). The remaining international border indicator, $BRDR_{INTL_{ij}}$, captures international trade between non-African countries (ROW–ROW trade), ensuring that AfCFTA-specific border effects are identified separately from other international trade relationships.

[5]The AfCFTA border indicator is constructed based on official signatory status. As documented in Table C.1, Eritrea is the only African country that has not signed the AfCFTA agreement, while the Sahrawi Arab Democratic Republic is not covered in the ITPD-E-R02 dataset. Consequently, the AfCFTA border is defined over the remaining 53 African countries.

The interpretation of the AfCFTA border coefficient in equation (5) differs depending on whether the border is treated as present in the observed data or removed in the counterfactual policy experiment. In the estimated gravity model, the coefficient β_3^u captures the average trade-cost wedge associated with international borders between AfCFTA members relative to domestic trade, conditional on all other trade-cost determinants and fixed effects. When the border is present, the percentage border effect is given by

$$\%\Delta_{BRDR_{AfCFTA_{ij}}\text{-present}} = (\exp(\beta_3^u) - 1) \times 100, \tag{6}$$

which measures the proportional reduction in bilateral trade attributable to the existence of the border. For example, a coefficient $\beta_3^u = -0.30$ implies that, all else equal, the presence of international borders reduces intra-AfCFTA trade by approximately 26% relative to domestic trade.

In the counterfactual analysis, the policy experiment consists of eliminating this border wedge by setting the AfCFTA border indicator to zero. Trade creation from border removal is therefore measured relative to the observed equilibrium in which the border is present. The relevant transformation is the inverse of the border effect and is given by:

$$\%\Delta_{BRDR_{AfCFTA_{ij}}\text{-removed}} = (\exp(-\beta_3^u) - 1) \times 100. \tag{7}$$

This expression captures the proportional increase in bilateral trade resulting from the removal of the border-induced trade cost. Continuing with the same example, a coefficient of -0.30 implies that eliminating the border between AfCFTA members would increase intra-AfCFTA trade by approximately 35%. The asymmetry between the border effect and the trade creation effect arises from the multiplicative structure of the gravity model: removing a trade barrier requires inverting the estimated trade-cost wedge rather than simply reversing its sign.[6]

Finally, the coefficient β_4^u captures border frictions affecting exports from AfCFTA members to the rest of the world. These frictions are not directly altered by AfCFTA implementation but are included to control for broader trends in international border effects, thereby ensuring that the estimated β_3^u isolates the AfCFTA-specific component of intra-AfCFTA border frictions.

3.3.2 Border-based General Equilibrium Welfare Analysis

To quantify the general equilibrium (GE) welfare effects of eliminating intra-AfCFTA border frictions under the AfCFTA, I implement a border-based counterfactual analysis using the General Equilibrium Poisson Pseudo–Maximum Likelihood (GEPPML) procedure proposed by Anderson et al. (2018). GEPPML provides a theoretically consistent and computationally tractable method for evaluating trade-policy counterfactuals within the structural gravity framework. Importantly, GEPPML is closely related to and operationalizes the exact-hat general equilibrium logic of Dekle et al. (2007, 2008) by recovering and updating multilateral resistance terms through PPML fixed effects rather than solving the nonlinear system explicitly.

[6] I refer the readers to Appendix B for the potential outcomes derivation of the AfCFTA border effect and border-removal trade creation.

The GE analysis proceeds in three steps: (i) baseline estimation, (ii) conditional general equilibrium adjustment holding nominal output and expenditure fixed, and (iii) full endowment general equilibrium adjustment allowing incomes and expenditures to respond endogenously to the policy shock. The GEPPML procedure described below is employed to implement the welfare estimates for each of the broad sectors in the ITPD-E-R02 trade dataset.[7]

Step 1: Baseline estimation I begin by estimating a sector-specific implementation of the structural gravity equation in equation (5) using cross-sectional data for 2019, incorporating both domestic and international trade flows:

$$X_{ij} = \exp\Big(\mathbf{T}_{ij}\boldsymbol{\beta} + \pi_i + \chi_j\Big) \times \varepsilon_{ij}, \tag{8}$$

where X_{ij} denotes exports from country i to country j (including $i = j$), π_i and χ_j are exporter and importer fixed effects, and $\boldsymbol{\beta}$ is a vector of coefficients. The bilateral trade-cost index $\mathbf{T}_{ij}\boldsymbol{\beta}$ collects standard gravity controls and border indicators and is specified as:

$$\mathbf{T}_{ij}\boldsymbol{\beta} = \eta \times \textbf{GRAVITY}_{ij} + \beta_1 RTA_{ij} + \beta_2 BRDR_{INTL_{ij}} + \beta_3 BRDR_{AfCFTA_{ij}} + \beta_4 BRDR_{AfCFTA \to ROW_{ij}}, \tag{9}$$

The AfCFTA border indicator $BRDR_{AfCFTA_{ij}}$ equals one for international trade between AfCFTA members and zero otherwise. Its coefficient, $\hat{\beta}_3$, captures the partial-equilibrium intra-African international border wedge relative to domestic trade. Baseline nominal output and expenditure are constructed as $Y_i = \sum_j X_{ij}$ and $E_j = \sum_i X_{ij}$, respectively.

Following Fally (2015) and Anderson et al. (2018), the estimated exporter and importer fixed effects implicitly recover the baseline outward and inward multilateral resistance (MR) terms. With Germany designated as the reference country (expenditure E_0), the MR terms are computed as:

$$OMR_i^0 = \frac{Y_i\, E_0}{\exp(\hat{\pi}_i)}, \qquad IMR_j^0 = \frac{E_j}{\exp(\hat{\chi}_j)\, E_0}. \tag{10}$$

Baseline real GDP, which serves as the welfare benchmark, is computed as $rGDP_i^0 = Y_i/(IMR_i^0)^{1/(1-\sigma)}$, where σ denotes the elasticity of substitution (set to 7 following standard practice).

Step 2: Conditional general equilibrium adjustment In the second step, I construct a counterfactual scenario that simulates full AfCFTA implementation by eliminating the intra-African international border wedge. Specifically, the AfCFTA border indicator is set to zero for all international AfCFTA member pairs, while all other bilateral trade-cost shifters are held fixed. This yields the counterfactual bilateral trade-cost index:

$$\mathbf{T}_{ij}^c\hat{\boldsymbol{\beta}} = \eta \times \textbf{GRAVITY}_{ij} + \beta_1 RTA_{ij} + \beta_2 BRDR_{INTL_{ij}} + \beta_4 BRDR_{AfCFTA \to ROW_{ij}}, \tag{11}$$

with domestic trade costs ($i = j$) unchanged.

[7] Appendix A provides the full description of the GEPPML implementation of AfCFTA border-removal welfare effects.

Conditional general equilibrium effects are obtained by re-estimating the gravity equation using PPML with the counterfactual trade-cost index imposed as an offset:

$$X_{ij} = \exp\Big(\mathbf{T}_{ij}^{c}\hat{\boldsymbol{\beta}} + \pi_i^c + \chi_j^c\Big) \times \varepsilon_{ij}^c, \tag{12}$$

where nominal output Y_i and expenditure E_j are held fixed at baseline levels, while exporter and importer fixed effects adjust to satisfy market-clearing conditions. The conditional MR terms are then computed as:

$$OMR_i^c = \frac{Y_i\,E_0}{\exp(\hat{\pi}_i^c)}, \qquad IMR_j^c = \frac{E_j}{\exp(\hat{\chi}_j^c)\,E_0}. \tag{13}$$

Conditional welfare changes are computed as the ratio of real incomes, which simplifies to the inverse ratio of inward MR terms:

$$W_i^C = \frac{rGDP_i^c}{rGDP_i^0} = \left(\frac{IMR_i^0}{IMR_i^c}\right)^{\frac{1}{1-\sigma}}, \tag{14}$$

capturing the price-index effect of the policy shock while holding nominal incomes fixed.

Step 3: Full endowment general equilibrium adjustment In the final step, nominal output and expenditure are allowed to respond endogenously to the trade-policy shock while maintaining constant trade-imbalance ratios $\phi_i \equiv E_i/Y_i$. Changes in exporter fixed effects between the baseline and conditional equilibria imply changes in factory-gate prices:

$$\frac{p_i^c}{p_i} = \left(\frac{\exp(\pi_i^c)}{\exp(\pi_i)}\right)^{\frac{1}{1-\sigma}}. \tag{15}$$

Nominal income and expenditure are updated proportionally to price changes, and bilateral trade flows are updated according to the structural gravity relationship:

$$\widetilde{X}_{ij}^{(m)} = \widehat{X}_{ij}^{(m-1)} \cdot \frac{\widehat{p}_i^{(m-1)}\,\widehat{p}_j^{(m-1)}}{\Delta OMR_i^{(m-1)}\,\Delta IMR_j^{(m-1)}}, \tag{16}$$

where $\Delta OMR_i^{(m-1)}$ and $\Delta IMR_j^{(m-1)}$ denote the change ratios in outward and inward MR terms from the previous iteration. This procedure mirrors the exact-hat updating logic of Dekle et al. (2007, 2008) and is implemented iteratively using PPML with offsets until convergence in factory-gate prices is achieved (see Appendix A for full implementation details).

Full endowment welfare effects are computed as changes in real GDP:

$$W_i^{FE} = \frac{rGDP_i^{FE}}{rGDP_i^0} = \frac{Y_i^{FE}/(IMR_i^{FE})^{1/(1-\sigma)}}{Y_i^0/(IMR_i^0)^{1/(1-\sigma)}}, \tag{17}$$

capturing both the direct gains from reduced intra-African trade costs and the indirect general equilibrium adjustments operating through factory-gate prices, incomes, expenditures, and multilateral resistance terms.

This GEPPML-based approach yields general equilibrium welfare estimates that are fully consistent with structural gravity theory and the exact-hat logic of Dekle et al. (2007, 2008), while allowing the AfCFTA border removal to be transparently mapped into a bilateral trade-cost shock and propagated through the global trading system.

4 Main findings & Discussions

4.1 Impacts of AfCFTA Agreement on Intra-African Trade

This section focuses on the interpretation and discussion of the empirical results of the impacts of the AfCFTA agreement on intra-African trade flows at the sectoral and industry level. Starting with the results of the sectoral analysis in Table 2, the gravity estimates cover the border effects of the AfCFTA agreement on total, agricultural, manufacturing, mining & energy, and services trade flows from columns (1) to (5), respectively. From Table 2, the structural gravity covariates have expected signs and statistical significance comparable to the existing estimates from sector-level gravity literature (see Borchert et al., 2022a; Larch et al., 2023). The coefficients of contiguity, common language, and colonial ties are positive and statistically significant for most of the sectors, indicating that countries with shared borders, common language, and historical colonial ties tend to trade significantly more than those without these shared characteristics. Conversely, distance has a negative and statistically significant effect on trade flows across all sectors, indicating that greater distance reduces trade. This is consistent with the gravity model of trade, which predicts that trade decreases with distance due to higher transportation costs and other trade barriers. For the RTA, it has a positive and statistically significant effect on total and sectoral trade, though the estimate for services is negligible. This suggests that membership in regional trade agreements enhances trade flows among member countries.

Proceeding to the estimates of the border variables, they capture the impact of observable and unobservable frictions that affect international relative to domestic trade after controlling for structural gravity covariates. These estimates reveal three key findings. First, the estimates are all negative, relatively large, and statistically significant at 1%, which is consistent with the existing literature on the home-bias and border effects on trade (see Anderson et al., 2018; Larch et al., 2023). Second, for the total and sectoral trade flows, the estimates of border effects faced by African exports to ROW ($BRDR_{AfCFTA \to ROW}$) are the largest, followed by the border effects faced by intra-African trade ($BRDR_{AfCFTA}$) and border barriers faced by the trade between the ROW (BRDR$_{INTL}$). Third, inter-sectoral comparison reveals that the largest observable and unobservable trade frictions that are not captured by the standard gravity covariates are present in services trade. This is consistent with the findings by Larch et al. (2023). This reflects the fact that services trade faces significant regulatory, informational, and institutional barriers that extend beyond the geographical and cultural factors captured by traditional gravity covariates.

The main variable of interest is $BRDR_{AfCFTA}$. By construction, it captures the impacts of observable and unobservable barriers that affect intra-African international trade relative to domestic sales after accounting for all gravity covariates. The estimates suggest heterogeneous effects across sectors and for total trade flows, with the largest effects observed in services (-30% s.e. 1.073) followed by agriculture (-26% s.e. 0.798), total trade (-21% s.e. 0.764), manufacturing (-16% s.e. 0.971), and mining and energy (-15% s.e. 1.896).[8] This means that the presence of borders between AfCFTA members reduces international trade between them by an average of 15% in the mining and energy sector to 30% in services. This large effect reflects the fact that underdeveloped regulatory frameworks and infrastructure deficits disproportionately affect intra-African services trade more than other sectors. This is relatively consistent with the findings of Borchert et al. (2022a), who find that services trade has the largest border effect, followed by mining & energy and agriculture, while manufacturing has the smallest border effect. Also, the large border effects in the agricultural sector align with the results from Chebbi et al. (2016) that agricultural trade faces extremely high border frictions compared to industrial (manufactured) goods in North Africa. However, the low border effects in mining within AfCFTA members suggest this sector benefits from established preferential policy treatment, making it the most integrated sector for intra-African trade.

In the context of an ex-ante evaluation of the AfCFTA, the estimated intra-AfCFTA border coefficients quantify the trade frictions that would be eliminated under full and effective implementation of the agreement. Accordingly, the relevant measure of trade creation is obtained from the border-removal transformation,$\%\Delta_{BRDR_{AfCFTA_{ij}}\text{-removed}} = (\exp(-\beta_3^u) - 1) \times 100$, which captures the proportional increase in bilateral trade flows that would occur once the AfCFTA-related border wedge is removed. Interpreted in this way, the results imply substantial potential trade creation effects across sectors. Specifically, the complete elimination of intra-AfCFTA border frictions is predicted to increase total intra-African trade by approximately 27%, with sectoral trade expanding by about 42% in services, 34% in agriculture, 20% in manufacturing, and 17% in mining and energy, ceteris paribus. These estimates indicate that AfCFTA implementation would have markedly heterogeneous impacts across sectors, with services and agriculture experiencing the largest trade expansion due to their currently high border-related frictions, while manufacturing and mining and energy record more moderate gains.

Importantly, these trade-creation magnitudes should be interpreted as upper-bound long-run effects under full and effective AfCFTA implementation, abstracting from transitional arrangements, partial compliance, and gradual liberalization schedules. Relative to existing studies, the estimated gains are smaller but arguably more realistic. For example, Abrego et al. (2019) project an 82% increase in intra-African trade flows, while Mevel and Karingi (2012) estimate an increase of 128.4%. The larger effects reported in these studies reflect assumptions of near-complete elimination of tariffs and selected non-tariff barriers, while capturing only a limited set of unobservable trade frictions. By contrast, the border-based approach employed here captures both observable and unobservable trade barriers in a reduced-form manner, even at the sectoral level, yielding more conservative but policy-relevant estimates. Notably, the estimated trade creation in services

[8] These values are calculated using this percentage border effect equation: $\%\Delta_{BRDR_{AfCFTA_{ij}}\text{-present}} = (\exp(\beta_3^u) - 1) \times 100$ and the standard errors are obtained using the Delta method.

(42%) is broadly consistent with the finding of Mevel and Karingi (2012) that services exhibit the smallest gains among sectors (31.9%) in tariff-based simulations. The identification of services as the sector with the largest trade creation potential in this study underscores the presence of substantial unobservable barriers, such as regulatory heterogeneity, recognition of qualifications, and restrictions on cross-border service provision, that are not captured by standard gravity covariates. This highlights a key advantage of the border-variable approach over tariff-based methodologies, which are inherently ill-suited to analyze services trade given the absence of tariffs. More broadly, the pronounced sectoral heterogeneity—where sectors facing the highest initial border frictions experience the largest gains from integration—is consistent with theoretical predictions and provides more granular policy insights than aggregate estimates such as those in Geda and Yimer (2023), who report overall trade creation effects ranging from 42.6% to 60.75% without sectoral decomposition.

Table 2: Gravity Estimates of the AfCFTA Border Effects on Intra-African Trade

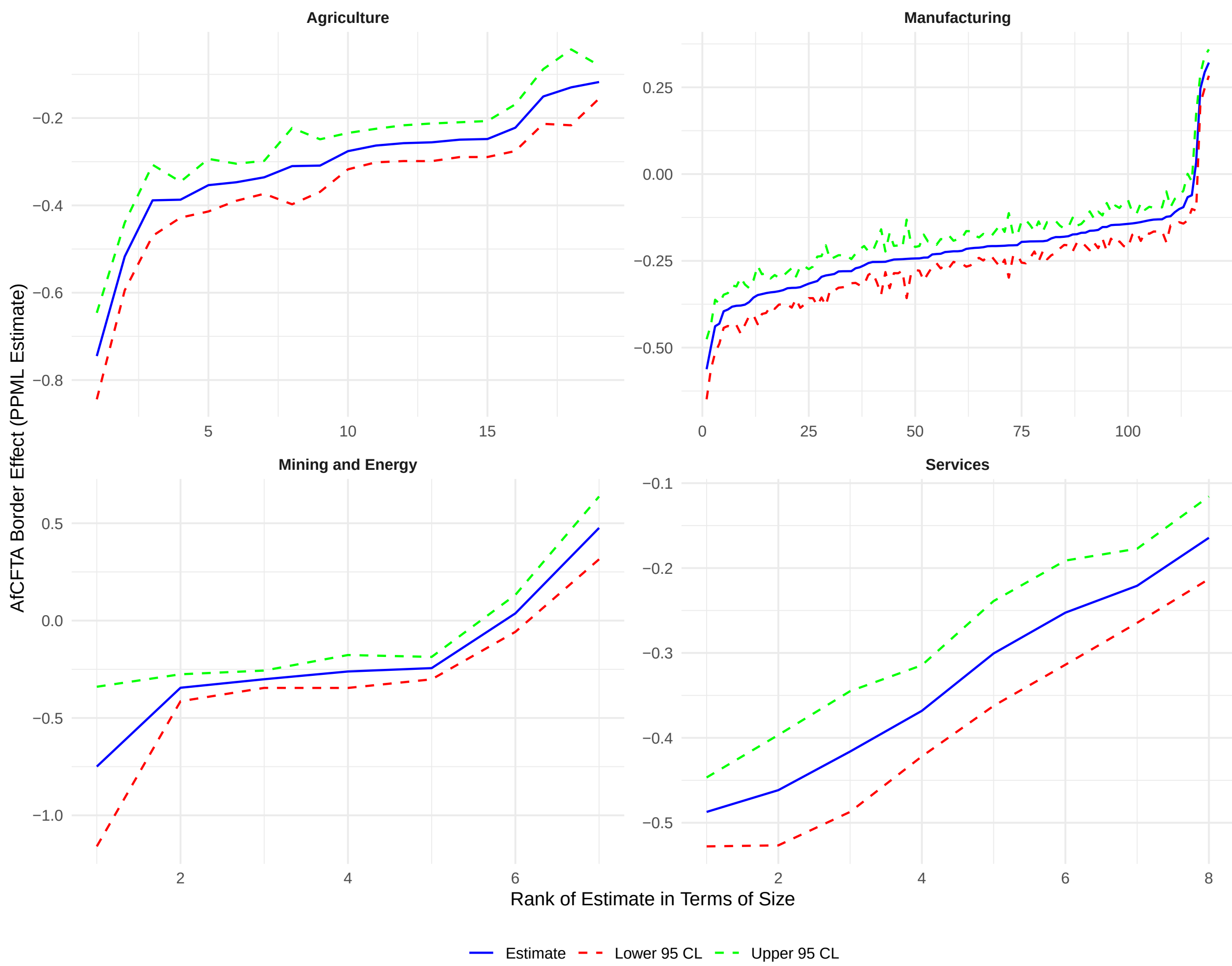
	(1) Total	(2) Agri	(3) Manu	(4) Mine	(5) Serv
CNTG	0.605 ***	0.534 ***	0.493 ***	0.412 *	0.532 ***
	(0.082)	(0.162)	(0.094)	(0.207)	(0.122)
COLN	0.350 ***	-0.002	0.227 *	0.554 *	0.633 ***
	(0.096)	(0.187)	(0.113)	(0.236)	(0.153)
DIST	-0.433 ***	-0.808 ***	-0.472 ***	-1.071 ***	-0.351 ***
	(0.046)	(0.072)	(0.053)	(0.090)	(0.054)
LANG	0.272 ***	0.272 *	0.311 ***	0.429 **	0.327 ***
	(0.056)	(0.110)	(0.069)	(0.131)	(0.094)
RTA	0.351 ***	0.896 ***	0.586 ***	0.446 ***	0.271 *
	(0.067)	(0.097)	(0.077)	(0.118)	(0.119)
BRDR_{INTL}	-0.197 ***	-0.189 ***	-0.131 ***	-0.195 ***	-0.258 ***
	(0.007)	(0.010)	(0.008)	(0.012)	(0.010)
BRDR_{AfCFTA}	-0.240 ***	-0.295 ***	-0.179 ***	-0.160 ***	-0.353 ***
	(0.010)	(0.011)	(0.012)	(0.022)	(0.015)
$\text{BRDR}_{AfCFTA\rightarrow ROW}$	-0.489 ***	-0.461 ***	-0.390 ***	-0.366 ***	-0.767 ***
	(0.015)	(0.022)	(0.018)	(0.034)	(0.024)
No of Observations	1929228	543247	879185	409238	97529

Notes: Statistics based on the author's calculations. The table presents the gravity estimates of the AfCFTA Border Effects on intra-African Total, Agriculture (Agri), Manufacturing (Manu), Mining & Energy (Mine), and Service (Serv) trade flows from columns (1) to (5), respectively. The data for each main sector is constructed by summing the data for all individual products within the corresponding main sector. The estimates are obtained using the PPML estimator with nominal bilateral trade flows (including zero trade flows), two-way (exporter-time and importer-time) fixed effects, and clustered standard errors (reported in parentheses) by country pair. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Next, I examine industry-level border effects for each of the 170 industries in the ITPD-E-R02 dataset. The industry-level estimates are plotted and grouped by broad sector in Figure 3, while the full set of estimates and their corresponding standard errors are reported in Table D.3. Figure 3 displays the estimated coefficients on the main variable of interest, $BRDR_{AfCFTA}$, for each industry. A small number of industry estimates are omitted due to collinearity in the border indicators arising from missing observations in some industries. The estimated border effects are predominantly negative and highly statistically significant, mostly at the 1% level. The magnitude of the estimated border effects varies substantially across industries, ranging from -0.061 for the *Other articles of paper and paperboard* industry in the manufacturing sector to -0.767 for the *Raw and refined sugar and sugar crops* industry in the agriculture sector. Interpreted as semi-elasticities, these estimates imply that the presence of international borders between AfCFTA members currently reduces industry-level intra-African trade by approximately 5.92% $\left[(\exp(-0.061) - 1) \times 100\right]$ to 53.56% $\left[(\exp(-0.767) - 1) \times 100\right]$, relative to domestic trade. The largest estimated border-related trade reductions are observed in *Raw and refined sugar and sugar crops* (53.56%), *Jewellery and related articles* (42.99%), *Eggs* (40.37%), *Dressing & dyeing of fur; processing of fur* (39.29%), and *Telecommunications, computer, and information services* (38.55%), spanning agriculture, manufacturing, and services. At the lower end of the distribution, the smallest border effects are recorded for *Basic chemicals except fertilizers* (10.42%), *Machinery for mining & construction* (9.61%), *Medical, surgical, and orthopaedic equipment* (9.06%), *Man-made fibres* (6.39%), and *Other articles of paper and paperboard* (5.92%), all of which belong to the manufacturing sector. These results highlight substantial heterogeneity in the intensity of border-related trade frictions across industries. Industries with larger estimated border effects are those for which international borders currently constitute a major impediment to cross-border exchange, whereas industries with smaller effects appear to be constrained more by production structure, scale economies, or technological factors than by border-related trade costs. Importantly, these estimates characterize the magnitude of existing border frictions; the corresponding trade creation effects from border removal are discussed separately using the appropriate counterfactual transformation.

While the preceding discussion focuses on the magnitude of the estimated border effects, it is equally important to translate these coefficients into trade creation effects implied by border removal under full AfCFTA implementation. The industry-level estimates imply substantially larger trade responses than those implied by the border-present effects alone. In particular, the elimination of intra-African border frictions is predicted to increase trade by approximately 115% in *Raw and refined sugar and sugar crops*, 75% in *Jewellery and related articles*, 68% in *Eggs*, 65% in *Dressing & dyeing of fur; processing of fur*, and 63% in *Telecommunications, computer, and information services*. These industries span agriculture, manufacturing, and services, underscoring that the largest trade creation effects are not confined to a single sector but arise where border-related frictions currently impose particularly severe constraints on cross-border exchange. At the lower end of the distribution, the smallest trade creation effects are observed in manufacturing industries, with predicted increases of 12% in *Basic chemicals except fertilizers*, 11% in *Machinery for mining & construction*, 10% in *Medical, surgical, and orthopaedic equipment*, 7% in *Man-made fibres*, and 6% in *Other articles of paper and paperboard*. The concentration of the lowest gains within manufacturing reflects the fact that trade in these industries is often driven more by scale economies, production networks, and technological intensity than by border-related trade costs alone. Consequently, while AfCFTA-induced reductions in border frictions do stimulate additional trade, the magnitude of the response is more limited than in sectors where borders constitute the dominant impediment to trade. These industry-level trade creation estimates reinforce the central message of the sectoral results: AfCFTA's long-run trade impacts are likely to be largest in activities characterized by high initial border frictions, particularly agriculture and selected services, while more modest but still positive gains are expected in manufacturing. By explicitly distinguishing between border effects and border-removal trade creation, these results provide a clearer mapping from estimated coefficients to economically meaningful counterfactual outcomes, and offer granular insights into which industries stand to benefit most from deeper continental integration.

Figure 3: Distribution of Border Effects of the AfCFTA at the Industry-level



Notes: Figure based on the author's calculations. The figure depicts the industry-level estimates (with the corresponding 95 percent confidence limits) of the AfCFTA border effects on intra-African trade covering Agriculture, Manufacturing, Mining and Energy, and Services sectors based on the ITPD-E-R02 trade dataset. These industry-level estimates refer to the coefficients of $BRDR_{AfCFTA_{ij}}$ in Eq. (5) ordered by magnitude (some dropped due to collinearity in border variables caused by missing observations in some industries).

4.2 Welfare Impacts of the AfCFTA Agreement

The welfare impacts of the AfCFTA agreement at the sector-level are reported in Table 3. The welfare estimates are the full endowment GE effects measured as percentage changes in real GDP, and correspond to the potential welfare benefits for abolishing borders between AfCFTA members. This analysis employs an elasticity of substitution of 7, consistent with established literature (see

Anderson et al., 2018; Larch et al., 2023). The table reports the welfare impacts for all AfCFTA signatories plus an aggregate Rest of World (ROW) for non-AfCFTA members, calculated as a weighted mean of all welfare changes using output shares as weights. The results demonstrate substantial heterogeneity across sectors and countries. The services sector in column *SERV* exhibits the most pronounced gains (up to 250% for Benin), followed by mining and energy in column *MINE* (up to 158% for Benin), agriculture in column *AGRI* (up to 93% for Botswana), and manufacturing in column *MANU* (up to 57% for Lesotho).

This sectoral variation reflects differences in pre-existing trade barriers and the potential for integration in regional value chains. A consistent pattern emerges across all sectors where smaller economies generally experience larger welfare gains than their larger counterparts. For instance, in the agricultural sector, Botswana (93.35%) and Lesotho (83.85%) demonstrate substantially higher gains than Egypt (12.91%) and Morocco (19.68%). Similarly, in manufacturing, Lesotho (57.03%) and Botswana (53.26%) show significantly greater improvements than South Africa (12.12%) and Egypt (15.16%). In the mining and energy sector, Benin (157.55%) and Niger (145.28%) exhibit exceptional gains compared to Nigeria (6.28%) and Angola (5.38%). The services sector follows this pattern with Benin (250.71%) and Cameroon (223.95%) experiencing much larger benefits than Egypt (29.03%) and Nigeria (6.82%). This inverse relationship between country size and welfare gains aligns with established evidence by Arkolakis et al. (2012), Anderson et al. (2018), Larch et al. (2023). Larger economies with naturally higher home market shares tend to gain proportionally less from trade liberalization, validating the economic principle that smaller countries benefit disproportionately when moving from restricted trade to liberalized arrangements. This occurs because smaller economies face more binding constraints from limited domestic market size before liberalization, and when borders are removed, these economies gain access to substantially larger markets, enabling economies of scale and specialization according to comparative advantage. Additionally, smaller economies can more effectively integrate into regional value chains, particularly in services where regulatory harmonization creates significant opportunities for cross-border provision. The negligible impact on the ROW (approximately 0.0%) across sectors underscores that the benefits remain predominantly concentrated within the AfCFTA region, highlighting the agreement's potential to foster intra-African trade without significant trade diversion effects.

Comparing these results with the existing AfCFTA literature reveals that GE estimation scenarios with tariff and, in some cases, some reductions in NTBs yield very low welfare effects, consistent with the findings of Larch et al. (2023) that tariff-based GE simulation produces low welfare gain compared to the border-based approach. For instance, the GE analysis by Abrego et al. (2019) based on tariff and NTB scenario projected that the expected welfare gain from AfCFTA would be around 2% to 4%. Meanwhile, the results from Fofack et al. (2021) reveal much lower estimates of 0.12% from full endowment and 0.29% from dynamic GE estimation.

Table 3: Welfare Effects of AfCFTA Agreement Across Different Sectors (% Change in Real GDP)

ISO-3	Country	AGRI (%)	MANU (%)	MINE (%)	SERV (%)
DZA	Algeria	44.28	28.07	16.40	192.34
AGO	Angola	62.48	35.48	5.38	195.59
BEN	Benin	43.22	39.96	157.55	250.71
BWA	Botswana	93.36	53.26	35.25	208.05
BFA	Burkina Faso	52.64	36.57	82.91	86.38
BDI	Burundi	28.24	42.06	87.36	193.22
CMR	Cameroon	36.02	36.46	63.91	223.95
CPV	Cape Verde	43.13	33.39	120.56	175.69
CAF	Central African Republic	73.93	41.44	140.77	191.88
TCD	Chad	74.25	40.00	57.78	214.24
COM	Comoros	18.08	36.46	114.32	183.71
CIV	Côte d'Ivoire	13.24	30.64	37.90	185.38
COD	Democratic Republic of the Congo	76.06	35.73	73.80	200.99
DJI	Djibouti	28.49	26.14	51.80	183.35
EGY	Egypt	12.91	15.16	19.64	29.03
GNQ	Equatorial Guinea	54.46	39.11	25.48	187.87
SWZ	Eswatini	63.73	44.32	88.21	84.18
ETH	Ethiopia	25.98	30.69	93.26	208.30
GAB	Gabon	75.68	38.63	23.35	193.60
GMB	Gambia	51.52	38.24	125.76	136.84
GHA	Ghana	22.48	26.23	24.73	193.80
GIN	Guinea	72.17	34.00	19.00	192.71
GNB	Guinea-Bissau	30.25	37.33	100.62	181.39
KEN	Kenya	21.19	26.76	55.58	40.40
LSO	Lesotho	83.85	57.03	57.00	138.93
LBR	Liberia	46.61	28.80	37.70	196.08
LBY	Libya	67.04	35.90	15.13	207.67
MDG	Madagascar	31.65	36.74	56.80	187.63
MWI	Malawi	38.61	44.54	133.44	194.30
MLI	Mali	31.54	32.58	135.78	205.11
MRT	Mauritania	68.02	35.50	39.42	190.54
MUS	Mauritius	20.86	23.70	23.73	184.41
MAR	Morocco	19.68	15.26	51.71	27.87
MOZ	Mozambique	54.18	45.71	40.88	192.19
NAM	Namibia	62.77	45.14	62.41	98.38
NER	Niger	61.56	40.52	145.28	126.55
NGA	Nigeria	36.72	24.50	6.28	6.82
COG	Republic of the Congo	44.57	35.94	21.49	191.73
RWA	Rwanda	29.53	36.21	47.86	200.20
SEN	Senegal	27.70	25.93	38.83	57.63
SYC	Seychelles	22.24	22.98	45.46	182.38
SLE	Sierra Leone	46.55	38.83	48.61	192.22
SOM	Somalia	59.04	37.31	105.06	205.79
ZAF	South Africa	24.12	12.12	18.30	35.11
SSD	South Sudan	79.87	41.47	39.69	210.92
SDN	Sudan	35.64	37.62	57.27	219.21
STP	São Tomé and Príncipe	52.35	39.61	142.70	187.79
TZA	Tanzania	49.77	36.34	68.30	219.10
TGO	Togo	49.20	39.03	64.57	199.16
TUN	Tunisia	33.29	22.87	73.73	172.54
UGA	Uganda	37.97	34.94	98.94	211.37
ZMB	Zambia	50.52	35.06	72.80	67.32
ZWE	Zimbabwe	41.50	48.94	68.11	209.75
ROW	Rest of World	-0.08	-0.06	-0.01	-0.00

Notes: Estimates are based on author's calculation following the GE PPML procedure prescribed by Anderson et al. (2018). This table reports the welfare effects (percentage change in real GDP) of the AfCFTA agreement corresponding to the elimination of borders between AfCFTA members across broad sectors, namely AGRI = Agriculture sector, MANU = Manufacturing sector, MINE = Mining and Energy sector, and SERV = Services sector, assuming an elasticity of substitution equals to 7. All estimates are based on ITPD-E-R02 data. ROW = Rest of World (weighted average for non-AfCFTA countries).

5 Robustness Checks

This section assesses the robustness of the partial- and general-equilibrium results to alternative sample constructions, parameter choices, and data sources.

Excluding zero trade flows. I first re-estimate the sectoral and industry-level models after excluding zero bilateral trade flows from the ITPD-E-R02 dataset. This exercise is motivated by the high prevalence of zero observations, particularly in services, and aims to assess whether large estimated border effects are driven by extensive-margin adjustments rather than intensive-margin trade responses. The results for total and sector-level trade are reported in Table D.4. Focusing on the coefficient of the AfCFTA border indicator, the estimated border effects remain close to the baseline results reported in Table 2. Specifically, the estimated semi-elasticities imply border-related trade reductions of approximately 21% for total trade, 25% for agriculture, 16% for manufacturing, 15% for mining and energy, and 27% for services. The persistence of the largest border effect in services is fully consistent with the main results and confirms that the findings are not driven by the presence of zero trade flows. While the primary focus of the robustness exercises is on border effects, it is informative to verify that the implied trade-creation magnitudes from border removal remain comparable. The robustness estimates imply trade creation of approximately 26% for total trade, 33% for agriculture, 20% for manufacturing, 17% for mining and energy, and 38% for services. These values are close to the baseline trade-creation estimates, indicating that the economic implications of AfCFTA border removal are not sensitive to the exclusion of zero trade flows. Industry-level estimates excluding zero trade flows are reported in Table D.5. The border effects range from -0.060 for *Other articles of paper and paperboard* to -0.589 for *Raw and refined sugar and sugar crops*, corresponding to trade reductions of approximately 6% to 45%. These industries coincide with the lower and upper bounds identified in the baseline analysis that includes zero trade flows, and the magnitudes are broadly similar to those reported in Table D.3. This confirms that the estimated heterogeneity in industry-level border effects is robust to the exclusion of zero observations.

Robustness to alternative elasticity of substitution. For the general-equilibrium analysis based on the ITPD-E-R02 dataset, I re-estimate welfare effects using a higher elasticity of substitution, $\sigma = 10$ and the estimates are reported in Table D.6. This sigma value lies within the range commonly used in the structural gravity literature and is particularly relevant for sector-level data, where substitution possibilities across varieties are likely to be stronger. As expected, the resulting welfare gains are slightly smaller in magnitude, consistent with the well-established result that higher elasticities dampen welfare responses to trade-cost reductions (Anderson and Van Wincoop, 2003; Yotov et al., 2016). Importantly, the ranking of sectors by welfare gains remains unchanged, confirming that the qualitative conclusions are not driven by the baseline choice of $\sigma = 7$.

Alternative data source and cross-dataset comparison. I further assess robustness using the Structural Gravity Dataset of Larch et al. (2019), which provides domestic and international manufacturing trade flows for 186 countries. Restricting the sample to 2000–2016, I re-estimate the partial-equilibrium model under alternative specifications that sequentially include WTO membership, preferential trade agreements, and economic integration agreements (Table D.7). Across all specifications, the estimated AfCFTA border effects are close to the baseline results, indicating robustness to alternative trade-policy controls and data sources. Using cross-sectional data for 2016, I then implement the GEPPML procedure on the Structural Gravity Dataset for different elasticities of substitution ($\sigma = 4, 5, 7$, and 10). The resulting welfare estimates are reported in Table D.8. Focusing on $\sigma = 7$, manufacturing welfare gains are somewhat larger than those obtained from the ITPD-E-R02 dataset, reflecting differences in country coverage, aggregation, and baseline trade structures. Nevertheless, the cross-country distribution of welfare gains is highly consistent across datasets, with countries experiencing relatively large (small) gains in one dataset exhibiting similar relative positions in the other. The additional estimates for alternative elasticities confirm the expected monotonic relationship between the elasticity of substitution and welfare magnitudes, providing further validation of the GE results. Finally, Figure D.1 visualizes the global distribution of welfare effects. The negligible welfare changes for non-AfCFTA members indicate that the gains from AfCFTA are largely concentrated within the region, suggesting limited trade diversion and reinforcing the agreement's potential to promote intra-African integration.

6 Conclusion and Policy Relevance

This paper conducts a comprehensive ex-ante evaluation of the African Continental Free Trade Area (AfCFTA) using a structural gravity framework that explicitly captures both observable and unobservable trade frictions through bilateral border indicators. By moving beyond tariff-based simulations, the analysis offers a more realistic assessment of the trade and welfare implications of deep continental integration, particularly in an environment where non-tariff and regulatory barriers remain central impediments to intra-African trade. The paper makes two principal contributions. First, it delivers a high degree of sectoral and industry-level granularity by estimating AfCFTA-related trade frictions across 170 industries spanning agriculture, manufacturing, mining and energy, and services. This level of disaggregation substantially extends the existing AfCFTA literature, which largely relies on aggregate or sector-level simulations, and provides policymakers with sharper guidance on where integration-induced trade gains are most likely to materialize. Second, the paper quantifies general equilibrium welfare effects using the General Equilibrium Poisson Pseudo–Maximum Likelihood (GEPPML) framework of Anderson et al. (2018). This approach operationalizes the exact-hat general equilibrium logic of Dekle et al. (2007, 2008) by recovering and iteratively updating multilateral resistance terms through PPML fixed effects, allowing for internally consistent welfare evaluation under counterfactual border removal.

The empirical results point to substantial but highly heterogeneous trade creation effects from eliminating intra-African border frictions. At the aggregate and sectoral levels, the removal of AfCFTA-related border wedges is estimated to increase intra-African trade by approximately 27% in total trade, with particularly large gains in services (42%) and agriculture (34%), followed by manufacturing (20%) and mining and energy (17%). These results indicate that sectors characterized by strong regulatory fragmentation and limited market access, especially services and agriculture, stand to benefit most from deeper continental integration. At the industry level, the analysis reveals even greater heterogeneity. Trade creation effects range from modest increases in capital- and scale-intensive manufacturing industries to very large gains in industries where border-related frictions currently play a dominant role. In particular, the evidence shows that full AfCFTA implementation could increase intra-African trade from approximately 6% to 115% at the industry-level. The welfare estimates show significant variation, with services exhibiting the largest gains (up to 250%), followed by mining and energy (up to 158%), agriculture (up to 93%), and manufacturing (up to 57%). Consistent with established economic theory, smaller economies generally experience larger welfare gains compared to their larger counterparts.

These findings carry important implications for AfCFTA implementation strategies and policy priorities. First, the substantial projected gains in services trade (42%) underscore the critical importance of advancing negotiations on services liberalization beyond the current focus on goods trade. Policymakers should prioritize harmonizing regulatory frameworks, mutual recognition agreements, and reducing administrative barriers that impede cross-border services provision. Second, the heterogeneous industry-level impacts suggest that countries should adopt differentiated approaches to implementation, focusing resources and capacity-building efforts on industries with the highest potential gains, such as sugar products, jewelry, and telecommunications services. Third, the finding that smaller economies stand to gain proportionally more from the agreement highlights the need for robust mechanisms to support these countries' integration efforts, including technical assistance for meeting RoO requirements and infrastructure development to facilitate market access. The significant role of non-tariff barriers revealed by this analysis, where border elimination effects substantially exceed tariff reduction scenarios from previous studies, emphasizes that successful AfCFTA implementation requires comprehensive reforms extending beyond tariff liberalization. Priority areas should include streamlining customs procedures, harmonizing standards and regulations, improving trade facilitation infrastructure, and addressing behind-the-border constraints. Furthermore, the projected welfare gains provide strong economic justification for accelerating RoO negotiations and implementation, which have delayed full trading under the agreement since 2021. Policymakers should consider pragmatic approaches to RoO flexibility, particularly for smaller and less-developed economies, to prevent these requirements from becoming prohibitive barriers to market access.

In conclusion, this comprehensive ex-ante evaluation demonstrates that the AfCFTA has significant potential to transform intra-African trade patterns and generate substantial welfare gains across member states, particularly for smaller economies. The projected economic benefits provide a compelling case for expediting the agreement's full implementation despite current challenges, particularly regarding RoO requirements. As African nations work to overcome these obstacles, the insights provided by this research can help guide strategic policy decisions to maximize the inclusive economic benefits of continent-wide integration. The AfCFTA represents a historic opportunity to advance Africa's economic development and regional integration goals as outlined in Agenda 2063, underscoring the importance of continued political commitment, pragmatic implementation strategies, and sustained research attention to ensure its successful realization.

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Appendix

A General Equilibrium PPML Implementation

This appendix provides a complete technical description of the general equilibrium (GE) procedure used to quantify the welfare effects of eliminating intra-African border frictions under the AfCFTA agreement. The implementation follows the General Equilibrium Poisson Pseudo–Maximum Likelihood (GEPPML) framework of Anderson et al. (2018), which operationalizes the exact-hat algebra of Dekle et al. (2007, 2008) by computing counterfactual equilibria in changes while recovering multilateral resistance terms from PPML fixed effects. The algorithm is implemented in `R` using the `fixest` package (Bergé, 2018) for high-dimensional fixed-effects PPML estimation.

Throughout, bilateral trade flows are denoted X_{ij}, where i indexes the exporter, j indexes the importer, and domestic trade corresponds to $i = j$. The elasticity of substitution is set to $\sigma = 7$, a standard calibration in the structural gravity literature (Head and Mayer, 2014; Yotov et al., 2016).

Notation and Variable Definitions. The following notation is used throughout this appendix. Let $CNTG_{ij}$, COL_{ij}, $LANG_{ij}$, and RTA_{ij} denote binary indicators for contiguity, colonial ties, common official language, and the presence of a regional trade agreement, respectively. The variable $DIST_{ij}$ denotes the logarithm of bilateral distance. Three mutually exclusive border indicators capture international trade frictions:

$$BRDR_{AfCFTA_{ij}} = \begin{cases} 1 & \text{if } i \in AfCFTA,\ j \in AfCFTA,\ i \neq j, \\ 0 & \text{otherwise,} \end{cases}$$

$$BRDR_{AfCFTA \to ROW_{ij}} = \begin{cases} 1 & \text{if } i \in AfCFTA,\ j \notin AfCFTA,\ i \neq j, \\ 0 & \text{otherwise,} \end{cases}$$

$$BRDR_{INTL_{ij}} = \begin{cases} 1 & \text{if } i \notin AfCFTA,\ j \notin AfCFTA,\ i \neq j, \\ 0 & \text{otherwise.} \end{cases}$$

In the empirical implementation, these correspond to the variables `afcfta_brdr`, `afcfta_row_brdr`, and `row_to_row`, respectively. Domestic trade observations ($i = j$) serve as the omitted reference category, so estimated border coefficients measure international trade frictions relative to intranational trade.

A.1 Baseline Scenario

A.1.1 Step 1.a: Baseline Gravity Estimation

The baseline equilibrium is characterized by estimating a structural gravity equation on cross-sectional data for 2019, including both domestic ($i = j$) and international ($i \neq j$) trade flows. Following Santos Silva and Tenreyro (2006) and Yotov et al. (2016), the gravity equation is estimated via PPML with exporter and importer fixed effects:

$$X_{ij} = \exp(\mathbf{T}_{ij}\boldsymbol{\beta} + \pi_i + \chi_j)\ \varepsilon_{ij}, \tag{A.1}$$

where π_i and χ_j denote exporter and importer fixed effects that control for multilateral resistance terms (Anderson and Van Wincoop, 2003; Feenstra, 2004), and ε_{ij} is an error term. The bilateral trade-cost index $\mathbf{T}_{ij}\boldsymbol{\beta}$ is specified as:

$$\mathbf{T}_{ij}\boldsymbol{\beta} = \beta_{\text{cntg}}CNTG_{ij} + \beta_{\text{col}}COL_{ij} + \beta_{\text{lang}}LANG_{ij} + \beta_{\text{dist}}DIST_{ij} + \beta_{\text{rta}}RTA_{ij} \\ + \beta_{A}\,BRDR_{AfCFTA_{ij}} + \beta_{AR}\,BRDR_{AfCFTA\to ROW_{ij}} + \beta_{RR}\,BRDR_{INTL_{ij}}. \tag{A.2}$$

Equation (A.1) is estimated using the `fepois` function from the `fixest` package (Bergé, 2018), which provides efficient PPML estimation with high-dimensional fixed effects.

A.1.2 Step 1.b: Baseline Trade Costs, Outputs, Expenditures, and Multilateral Resistances

Baseline outputs and expenditures. Baseline nominal output and expenditure are constructed from observed trade flows as:

$$Y_i \equiv \sum_j X_{ij}, \qquad E_j \equiv \sum_i X_{ij}.$$

For normalization, Germany (ISO code `DEU`) is used as the reference importer and is recoded as `ZZZ` in the data. Let E_0 denote the expenditure of the reference importer (Germany) in the baseline year.

Trade-cost index. The multiplicative bilateral trade-cost index is constructed as

$$t_{ij} \;\equiv\; \exp\Big(\mathbf{T}_{ij}\hat{\boldsymbol{\beta}}\Big). \tag{A.3}$$

Following Anderson et al. (2018), this object corresponds to the power-transformed trade-cost term:

$$t_{ij} = \tau_{ij}^{1-\sigma}, \qquad \text{so that} \qquad \ln t_{ij} = \mathbf{T}_{ij}\hat{\boldsymbol{\beta}}, \tag{A.4}$$

where $\tau_{ij} > 1$ represents the iceberg trade cost. This formulation is computationally convenient because the PPML estimation directly yields $\hat{\boldsymbol{\beta}}$, from which t_{ij} is recovered without requiring separate calibration of τ_{ij}.

Baseline multilateral resistances from fixed effects. Exporter and importer fixed effects from (A.1) are used to recover outward and inward multilateral resistance (MR) terms. Following Anderson et al. (2018), with Germany designated as the reference country (with expenditure E_0), the MR terms are computed as:

$$OMR_i^0 = \frac{Y_i\,E_0}{\exp(\hat{\pi}_i)}, \tag{A.5}$$

$$IMR_j^0 = \frac{E_j}{\exp(\hat{\chi}_j)\,E_0}. \tag{A.6}$$

Baseline real GDP. Baseline real GDP, which serves as the welfare metric, is computed as:

$$rGDP_i^0 = \frac{Y_i}{\left(IMR_i^0\right)^{\frac{1}{1-\sigma}}}. \tag{A.7}$$

This expression follows from the structural gravity framework, where real income equals nominal income deflated by the consumer price index, which is a monotonic transformation of the inward multilateral resistance term (Anderson and Van Wincoop, 2003; Arkolakis et al., 2012).

A.2 Conditional General Equilibrium Scenario

A.2.1 Step 2: Counterfactual Shock and Conditional PPML Estimation

Counterfactual shock. The AfCFTA policy experiment simulates the elimination of intra-African international border frictions by setting:

$$BRDR^c_{AfCFTA_{ij}} = 0 \quad \text{for} \quad i \neq j \text{ with } i \in AfCFTA,\ j \in AfCFTA,$$

while all other bilateral trade-cost determinants remain unchanged. Crucially, domestic trade costs are held fixed at baseline values, i.e., $t^c_{ii} = t^0_{ii}$ for all i. This assumption ensures that the counterfactual captures only the direct effect of removing international border frictions between AfCFTA members.

Counterfactual trade-cost index. The counterfactual multiplicative trade-cost index is defined as:

$$t^c_{ij} \;\equiv\; \exp\Big(\mathbf{T}^c_{ij}\hat{\boldsymbol{\beta}}\Big), \qquad \ln t^c_{ij} = \mathbf{T}^c_{ij}\hat{\boldsymbol{\beta}}, \tag{A.8}$$

where $\mathbf{T}^c_{ij}$ equals the baseline covariate vector $\mathbf{T}_{ij}$ except that $BRDR_{AfCFTA_{ij}}$ is set to zero for all AfCFTA member pairs $(i, j \in AfCFTA,\ i \neq j)$. This counterfactual represents the complete elimination of the intra-African border wedge.

Conditional PPML with offset. Conditional GE effects are obtained by re-estimating PPML with exporter and importer fixed effects, using the counterfactual log-index $\ln t^c_{ij}$ as an offset term:

$$X_{ij} = \exp\big(\pi^c_i + \chi^c_j + \ln t^c_{ij}\big)\ \varepsilon^c_{ij}. \tag{A.9}$$

The offset specification ensures that the counterfactual trade costs t^c_{ij} are imposed as known values rather than estimated, while the fixed effects (π^c_i, χ^c_j) adjust to satisfy the market-clearing conditions. In this conditional scenario, nominal outputs Y_i and expenditures E_j are held at their baseline values, allowing only the multilateral resistance terms to respond to the policy shock.

Conditional multilateral resistances and welfare. Using the fixed effects from (A.9), the conditional MR terms are computed as:

$$OMR^c_i = \frac{Y_i\, E_0}{\exp(\hat{\pi}^c_i)}, \tag{A.10}$$

$$IMR^c_j = \frac{E_j}{\exp(\hat{\chi}^c_j)\, E_0}, \tag{A.11}$$

Conditional real GDP is then computed as:

$$rGDP^c_i = \frac{Y_i}{(IMR^c_i)^{\frac{1}{1-\sigma}}}. \tag{A.12}$$

Conditional welfare effects are measured as the proportional change in real GDP:

$$W_i^C = \frac{rGDP_i^c}{rGDP_i^0} = \left(\frac{IMR_i^0}{IMR_i^c}\right)^{\frac{1}{1-\sigma}}. \tag{A.13}$$

The conditional welfare measure captures the immediate price-index effects of the policy shock while holding nominal incomes fixed. It represents a short-run adjustment in which production patterns have not yet responded to the new trade environment.

A.3 Full Endowment General Equilibrium

A.3.1 Step 3: Full Endowment Iteration

In the full endowment equilibrium, nominal outputs and expenditures are permitted to adjust endogenously in response to the policy shock. Following Anderson et al. (2018), the trade-imbalance ratio for each country is held constant:

$$\phi_i \equiv \frac{E_i}{Y_i}, \qquad \text{with } \phi_i \text{ fixed at baseline values throughout the counterfactual.} \tag{A.14}$$

This assumption, standard in the GE gravity literature (Dekle et al., 2007; Caliendo and Parro, 2015), ensures that the model is closed by requiring aggregate expenditure to track changes in aggregate income proportionally. The parameter ϕ_i is computed from domestic observations ($i = j$) in the baseline data.

A.3.2 Step 3.a: Initialization

Let $\widehat{X}_{ij}^c$ denote the fitted trade flows from the conditional PPML regression (A.9). The iterative procedure is initialized by setting:

$$\widehat{X}_{ij}^{(1)} \leftarrow \widehat{X}_{ij}^c.$$

Let $\exp(\hat{\pi}_i^0)$ and $\exp(\hat{\chi}_j^0)$ denote the exponentiated exporter and importer fixed effects from the baseline regression (A.1), and let $\exp(\hat{\pi}_i^c)$ and $\exp(\hat{\chi}_j^c)$ denote those from the conditional regression (A.9). Initial factory-gate price changes are computed from the ratio of conditional to baseline exporter fixed effects:

$$\widehat{p}_i^{(1)} \equiv \left(\frac{\exp(\hat{\pi}_i^c)}{\exp(\hat{\pi}_i^0)}\right)^{\frac{1}{1-\sigma}}. \tag{A.15}$$

An importer-side price analog is constructed using the average of exporter fixed effects computed on domestic observations, grouped by importer:

$$\widehat{p}_{j,\text{imp}}^{(1)} \equiv \left(\frac{\overline{\exp(\hat{\pi}^c)}_j}{\overline{\exp(\hat{\pi}^0)}_j}\right)^{\frac{1}{1-\sigma}}, \tag{A.16}$$

where $\overline{\exp(\hat{\pi})}_j$ denotes the mean of $\exp(\hat{\pi}_i)$ computed over observations with $i = j$, grouped by importer j. This construction exploits the fact that on the diagonal ($i = j$), the exporter fixed effect reflects the domestic factory-gate price.

A.3.3 Step 3.b: Iterative Procedure

For iteration index $m = 2, 3, \ldots$, the algorithm repeats the following sequence of updates until convergence is achieved.

(i) Trade update. Define the change ratios in outward and inward multilateral resistances from the previous iteration:

$$\Delta OMR_i^{(m-1)} \equiv \frac{OMR_i^{(m-1)}}{OMR_i^{(m-2)}}, \qquad \Delta IMR_j^{(m-1)} \equiv \frac{IMR_j^{(m-1)}}{IMR_j^{(m-2)}}.$$

The trade flows used as the dependent variable in the next PPML iteration are updated according to:

$$X_{ij}^{(m)} = \widehat{X}_{ij}^{(m-1)} \cdot \frac{\widehat{p}_i^{(m-1)}\, \widehat{p}_{j,\text{imp}}^{(m-1)}}{\Delta OMR_i^{(m-1)}\, \Delta IMR_j^{(m-1)}}. \tag{A.17}$$

This update formula corresponds to Equation (14) in Anderson et al. (2018) and reflects the structural gravity relationship linking bilateral trade to factory-gate prices and multilateral resistance terms.

(ii) PPML re-estimation with offset. Using $X_{ij}^{(m)}$ from (A.17) as the dependent variable, PPML is re-estimated with exporter and importer fixed effects and with the counterfactual trade-cost offset:

$$X_{ij}^{(m)} = \exp\left(\pi_i^{(m)} + \chi_j^{(m)} + \ln t_{ij}^c\right) \varepsilon_{ij}^{(m)}. \tag{A.18}$$

Let $\widehat{X}_{ij}^{(m)}$ denote the fitted values from this regression.

(iii) Update nominal outputs, expenditures, and reference expenditure. Nominal outputs and expenditures are updated from fitted trade flows:

$$Y_i^{(m)} \equiv \sum_j \widehat{X}_{ij}^{(m)}, \qquad E_j^{(m)} \equiv \sum_i \widehat{X}_{ij}^{(m)}. \tag{A.19}$$

The reference expenditure $E_0^{(m)}$ is the expenditure of Germany (the reference country) computed from (A.19).

(iv) Update multilateral resistances. Using the newly estimated fixed effects from (A.18), the multilateral resistance terms are updated as:

$$OMR_i^{(m)} \equiv \frac{Y_i^{(m)}}{\exp(\hat{\pi}_i^{(m)})}, \tag{A.20}$$

$$IMR_j^{(m)} \equiv \frac{E_j^{(m)}}{\exp(\hat{\chi}_j^{(m)})\, E_0^{(m)}}. \tag{A.21}$$

The change ratios $\Delta OMR_i^{(m)}$ and $\Delta IMR_j^{(m)}$ for the next iteration are computed from (A.20)–(A.21).

(v) Update factory-gate prices. Exporter-side factory-gate prices are updated using changes in exporter fixed effects, adjusted by changes in the reference expenditure:

$$\widehat{p}_i^{(m)} = \left[\frac{\exp(\hat{\pi}_i^{(m)})}{\exp(\hat{\pi}_i^{(m-1)})} \Big/ \frac{E_0^{(m)}}{E_0^{(m-1)}} \right]^{\frac{1}{1-\sigma}}. \tag{A.22}$$

Importer-side price changes are updated analogously using importer-specific averages of exporter fixed effects on domestic observations:

$$\widehat{p}_{j,\text{imp}}^{(m)} = \left[\frac{\overline{\exp(\hat{\pi}^{(m)})}_j}{\overline{\exp(\hat{\pi}^{(m-1)})}_j} \Big/ \frac{E_0^{(m)}}{E_0^{(m-1)}} \right]^{\frac{1}{1-\sigma}}. \tag{A.23}$$

The normalization by the reference expenditure ratio ensures that price changes are measured relative to the numéraire country.

Convergence criterion. The iteration terminates when the change in exporter-side prices between successive iterations is sufficiently small. Specifically, convergence is declared when both the standard deviation and the maximum absolute difference of $\widehat{p}_i^{(m)} - \widehat{p}_i^{(m-1)}$ across all exporters fall below a tolerance of 0.001. A maximum of 30 iterations is imposed as a safeguard against non-convergence.

A.3.4 Step 3.c: Full Endowment GE Indexes and Welfare

Upon convergence, the full endowment nominal output is constructed using the converged factory-gate price factor and baseline output:

$$Y_i^{FE} \equiv \widehat{p}_i^{FE}\, Y_i^0, \tag{A.24}$$

where $\widehat{p}_i^{FE}$ denotes the converged factory-gate price. Expenditures are updated using the constant trade-imbalance ratio:

$$E_j^{FE} \equiv \phi_j\, Y_j^{FE}. \tag{A.25}$$

The final multilateral resistance terms are computed using the converged fixed effects and the reference expenditure E_0^{FE}:

$$OMR_i^{FE} = \frac{Y_i^{FE}\, E_0^{FE}}{\exp(\hat{\pi}_i^{FE})}, \tag{A.26}$$

$$IMR_j^{FE} = \frac{E_j^{FE}}{\exp(\hat{\chi}_j^{FE})\, E_0^{FE}}. \tag{A.27}$$

Full endowment real GDP is computed as:

$$rGDP_i^{FE} = \frac{Y_i^{FE}}{\left(IMR_i^{FE}\right)^{\frac{1}{1-\sigma}}}. \tag{A.28}$$

Finally, full endowment welfare changes are reported as the proportional change in real GDP:

$$W_i^{FE} = \frac{rGDP_i^{FE}}{rGDP_i^0}. \tag{A.29}$$

Summary. This GEPPML implementation is fully consistent with the exact-hat algebra of Dekle et al. (2007, 2008) and follows the three-step estimation procedure of Anderson et al. (2018). The baseline scenario recovers initial multilateral resistances from PPML fixed effects; the conditional scenario imposes the counterfactual trade costs as an offset while holding nominal incomes fixed; and the full endowment scenario iterates to convergence, allowing outputs and expenditures to adjust endogenously subject to balanced trade-imbalance ratios. The welfare effects reported in the main text are computed as percentage changes in real GDP between the full endowment counterfactual and the baseline.

B Potential-Outcomes Derivation of the AfCFTA Border Effect and Border-Removal Trade Creation

This appendix derives, using a simple potential-outcomes framework, the percentage change in bilateral trade associated with (i) the *presence* of an AfCFTA international border and (ii) the *removal* of that border in the counterfactual analysis. The derivation applies directly to the AfCFTA border indicator used in the PPML structural gravity estimation.

B.1 Potential outcomes and the AfCFTA border indicator

For each exporter–importer pair (i, j) and year t, let D^A_{ij} denote the AfCFTA border indicator:

$$D^A_{ij} = \begin{cases} 1 & \text{if } i \text{ and } j \text{ are AfCFTA members and } i \neq j \text{ (international intra-AfCFTA trade)}, \\ 0 & \text{otherwise.} \end{cases}$$

Let $X^u_{ij,t}$ denote observed trade flows at unit of analysis u (sector or industry). In the potential-outcomes framework, each (i, j, t) has two potential trade outcomes:

$$X^u_{ij,t}(1) \quad \text{and} \quad X^u_{ij,t}(0),$$

where $X^u_{ij,t}(1)$ is the trade flow that would be observed if an AfCFTA border is present ($D^A_{ij} = 1$) and $X^u_{ij,t}(0)$ is the trade flow that would be observed if the AfCFTA border is absent ($D^A_{ij} = 0$). The observed outcome satisfies

$$X^u_{ij,t} = D^A_{ij} X^u_{ij,t}(1) + \left(1 - D^A_{ij}\right) X^u_{ij,t}(0).$$

B.2 PPML structural gravity mean and implied potential outcomes

Consider the PPML gravity specification in which the AfCFTA border indicator enters additively in the index:

$$\mathbb{E}\left[X^u_{ij,t} \mid Z^u_{ij,t}, D^A_{ij}\right] = \exp\left(\mu^u_{ij,t} + \beta^u_3 D^A_{ij}\right), \tag{B.1}$$

where $\mu^u_{ij,t}$ collects all other components of the index (e.g., exporter–time and importer–time fixed effects, gravity covariates, RTAs, and other border indicators), and β^u_3 is the coefficient on the AfCFTA border indicator. Under (B.1), the conditional expectations of the two potential outcomes are

$$\mathbb{E}\left[X^u_{ij,t}(1) \mid Z^u_{ij,t}\right] = \exp\left(\mu^u_{ij,t} + \beta^u_3\right), \tag{B.2}$$

$$\mathbb{E}\left[X^u_{ij,t}(0) \mid Z^u_{ij,t}\right] = \exp\left(\mu^u_{ij,t}\right). \tag{B.3}$$

B.3 Percentage change when the AfCFTA border is present

Define the percentage difference in expected trade when the AfCFTA border is present relative to the no-border potential outcome as:

$$\%\Delta^u_{\text{border\_present}} \equiv \frac{\mathbb{E}\left[X^u_{ij,t}(1) \mid Z^u_{ij,t}\right] - \mathbb{E}\left[X^u_{ij,t}(0) \mid Z^u_{ij,t}\right]}{\mathbb{E}\left[X^u_{ij,t}(0) \mid Z^u_{ij,t}\right]} \times 100.$$

Substituting (B.2)–(B.3) yields

$$\begin{aligned}\%\Delta^u_{\text{border\_present}} &= \frac{\exp(\mu^u_{ij,t} + \beta^u_3) - \exp(\mu^u_{ij,t})}{\exp(\mu^u_{ij,t})} \times 100 \\ &= (\exp(\beta^u_3) - 1) \times 100.\end{aligned}$$

Hence, in the PPML gravity model, the AfCFTA border coefficient β^u_3 implies that the presence of an AfCFTA international border changes trade by $(\exp(\beta^u_3) - 1) \times 100$ percent relative to the no-border benchmark.

B.4 Percentage change from removing the AfCFTA border (trade creation relative to the status quo)

In the counterfactual analysis relevant for AfCFTA implementation, the status quo for intra-AfCFTA international trade corresponds to $D^A_{ij} = 1$. Trade creation from removing the border compares the no-border potential outcome to the current border-present outcome. Define:

$$\%\Delta^u_{\text{border\_removed}} \equiv \frac{\mathbb{E}\left[X^u_{ij,t}(0) \mid Z^u_{ij,t}\right] - \mathbb{E}\left[X^u_{ij,t}(1) \mid Z^u_{ij,t}\right]}{\mathbb{E}\left[X^u_{ij,t}(1) \mid Z^u_{ij,t}\right]} \times 100.$$

Substituting (B.2)–(B.3) yields

$$\begin{aligned}\%\Delta^u_{\text{border\_removed}} &= \frac{\exp(\mu^u_{ij,t}) - \exp(\mu^u_{ij,t} + \beta^u_3)}{\exp(\mu^u_{ij,t} + \beta^u_3)} \times 100 \\ &= (\exp(-\beta^u_3) - 1) \times 100.\end{aligned}$$

Therefore, the counterfactual trade creation effect of eliminating the AfCFTA border wedge, relative to the border-present status quo, is $(\exp(-\beta^u_3) - 1) \times 100$ percent.

The two percentage changes differ because they use different baselines. The "border-present" effect expresses $X^u_{ij,t}(1)$ relative to $X^u_{ij,t}(0)$, whereas the "border-removed" effect expresses $X^u_{ij,t}(0)$ relative to $X^u_{ij,t}(1)$. With a multiplicative PPML mean, reversing the comparison requires taking the inverse ratio, which yields $\exp(-\beta^u_3)$ rather than $\exp(\beta^u_3)$.

C AfCFTA Timeline & Ratification Status

Table C.1: AfCFTA Ratification Status of African Countries

State	Signed Agreement	Ratification Date	State	Signed Agreement	Ratification Date
Algeria	YES	23/06/2021	Libya	YES	—
Angola	YES	04/11/2020	Madagascar	YES	—
Benin	YES	—	Malawi	YES	15/01/2021
Botswana	YES	19/02/2023	Mali	YES	01/02/2019
Burkina Faso	YES	29/05/2019	Mauritania	YES	11/02/2019
Burundi	YES	26/08/2021	Mauritius	YES	07/10/2019
Cape Verde	YES	05/02/2022	Morocco	YES	20/04/2022
Cameroon	YES	01/12/2020	Mozambique	YES	05/07/2023
Central African Rep.	YES	22/09/2020	Namibia	YES	01/02/2019
Chad	YES	02/07/2018	Niger	YES	19/06/2018
Comoros	YES	19/02/2023	Nigeria	YES	05/12/2020
Congo, Dem. Rep.	YES	23/02/2022	Rwanda	YES	26/05/2018
Congo, Rep.	YES	10/02/2019	Sahrawi Arab Democratic Republic	YES	30/04/2019
Côte d'Ivoire	YES	23/11/2018	São Tomé and Príncipe	YES	27/06/2019
Djibouti	YES	11/02/2019	Senegal	YES	02/04/2019
Egypt	YES	08/04/2019	Seychelles	YES	15/09/2021
Equatorial Guinea	YES	02/07/2019	Sierra Leone	YES	30/04/2019
Eritrea	NO	—	Somalia	YES	—
Eswatini	YES	02/07/2018	South Africa	YES	10/02/2019
Ethiopia	YES	10/04/2019	South Sudan	YES	—
Gabon	YES	07/07/2019	Sudan	YES	—
Gambia	YES	16/04/2019	Tanzania	YES	17/01/2022
Ghana	YES	10/05/2018	Togo	YES	02/04/2019
Guinea	YES	16/10/2018	Tunisia	YES	27/11/2020
Guinea/Bissau	YES	27/09/2022	Uganda	YES	09/02/2019
Kenya	YES	10/05/2018	Zambia	YES	05/02/2021
Lesotho	YES	27/11/2020	Zimbabwe	YES	24/05/2019
Liberia	YES	31/07/2024			

Note: Information on the signatory States and their ratification dates is gathered from Tralac (2025).

D Additional Results

Table D.2: List of countries in the general equilibrium analysis of the AfCFTA agreement

Country	Country	Country	Country
Albania (ALB)	Dominican Republic (DOM)	Lao PDR (LAO)	Romania (ROU)
Algeria (DZA)	Ecuador (ECU)	Latvia (LVA)	Russia (RUS)
Angola (AGO)	Egypt (EGY)	Lebanon (LBN)	Rwanda (RWA)
Argentina (ARG)	El Salvador (SLV)	Lesotho (LSO)	São Tomé and Príncipe (STP)
Armenia (ARM)	Equatorial Guinea (GNQ)	Liberia (LBR)	Saudi Arabia (SAU)
Australia (AUS)	Estonia (EST)	Libya (LBY)	Senegal (SEN)
Austria (AUT)	Eswatini (SWZ)	Lithuania (LTU)	Serbia (SRB)
Azerbaijan (AZE)	Ethiopia (ETH)	Luxembourg (LUX)	Seychelles (SYC)
Bahamas (BHS)	Fiji (FJI)	Madagascar (MDG)	Sierra Leone (SLE)
Bahrain (BHR)	Finland (FIN)	Malawi (MWI)	Singapore (SGP)
Bangladesh (BGD)	France (FRA)	Malaysia (MYS)	Slovakia (SVK)
Barbados (BRB)	Gabon (GAB)	Mali (MLI)	Slovenia (SVN)
Belarus (BLR)	Gambia (GMB)	Malta (MLT)	Somalia (SOM)
Belgium (BEL)	Georgia (GEO)	Mauritania (MRT)	South Africa (ZAF)
Benin (BEN)	Germany (DEU)	Mauritius (MUS)	South Sudan (SSD)
Bosnia and Herzegovina (BIH)	Ghana (GHA)	Mexico (MEX)	Spain (ESP)
Botswana (BWA)	Greece (GRC)	Moldova (MDA)	Sri Lanka (LKA)
Brazil (BRA)	Guatemala (GTM)	Montenegro (MNE)	Sudan (SDN)
Brunei (BRN)	Guinea (GIN)	Morocco (MAR)	Sweden (SWE)
Bulgaria (BGR)	Guinea-Bissau (GNB)	Mozambique (MOZ)	Switzerland (CHE)
Burkina Faso (BFA)	Honduras (HND)	Myanmar (MMR)	Taiwan (TWN)
Burundi (BDI)	Hong Kong (HKG)	Namibia (NAM)	Tajikistan (TJK)
Cabo Verde (CPV)	Hungary (HUN)	Nepal (NPL)	Tanzania (TZA)
Cambodia (KHM)	Iceland (ISL)	Netherlands (NLD)	Thailand (THA)
Cameroon (CMR)	India (IND)	New Zealand (NZL)	Togo (TGO)
Canada (CAN)	Indonesia (IDN)	Nicaragua (NIC)	Trinidad and Tobago (TTO)
Central African Republic (CAF)	Iran (IRN)	Niger (NER)	Tunisia (TUN)
Chad (TCD)	Iraq (IRQ)	Nigeria (NGA)	Turkey (TUR)
Chile (CHL)	Ireland (IRL)	North Macedonia (MKD)	Turkmenistan (TKM)
China (CHN)	Israel (ISR)	Norway (NOR)	Uganda (UGA)
Colombia (COL)	Italy (ITA)	Oman (OMN)	Ukraine (UKR)
Comoros (COM)	Jamaica (JAM)	Pakistan (PAK)	United Arab Emirates (ARE)
Congo, Democratic Republic (COD)	Japan (JPN)	Panama (PAN)	United Kingdom (GBR)
Congo, Republic (COG)	Jordan (JOR)	Papua New Guinea (PNG)	United States (USA)
Costa Rica (CRI)	Kazakhstan (KAZ)	Peru (PER)	Uzbekistan (UZB)
Côte d'Ivoire (CIV)	Kenya (KEN)	Philippines (PHL)	Venezuela (VEN)
Croatia (HRV)	Korea, South (KOR)	Poland (POL)	Vietnam (VNM)
Cuba (CUB)	Kuwait (KWT)	Portugal (PRT)	Yemen (YEM)
Cyprus (CYP)	Kyrgyzstan (KGZ)	Qatar (QAT)	Zambia (ZMB)
Czech Republic (CZE)			Zimbabwe (ZWE)
Denmark (DNK)			
Djibouti (DJI)			

Table D.3: Distribution of Border Effect of the AfCFTA at the Industry-level

ID	Industry Description	Broad Sector	Estimate	Std. Err.
1	Wheat	Agriculture	-0.222***	0.027
2	Rice (raw)	Agriculture	-0.256***	0.022
3	Corn	Agriculture	-0.258***	0.021
4	Other cereals	Agriculture	-0.276***	0.021
6	Soybeans	Agriculture	-0.310***	0.045
7	Other oilseeds (excluding peanuts)	Agriculture	-0.336***	0.019
9	Raw and refined sugar and sugar crops	Agriculture	-0.767***	0.049
10	Other sweeteners	Agriculture	-0.354***	0.031
11	Pulses and legumes, dried, preserved	Agriculture	-0.248***	0.021
12	Fresh fruit	Agriculture	-0.347***	0.022
13	Fresh vegetables	Agriculture	-0.387***	0.021
16	Nuts	Agriculture	-0.309***	0.031
19	Eggs	Agriculture	-0.517***	0.040
20	Other meats, livestock products, and live animals	Agriculture	-0.151***	0.032
21	Cocoa and cocoa products	Agriculture	-0.388***	0.042
22	Beverages, nec	Agriculture	-0.130***	0.044
24	Tobacco leaves and cigarettes	Agriculture	-0.250***	0.020
25	Spices	Agriculture	-0.263***	0.020
26	Other agricultural products, nec	Agriculture	-0.117***	0.019
29	Mining of hard coal	Mining & Energy	0.037	0.048
30	Mining of lignite	Mining & Energy	0.476***	0.082
31	Extraction crude petroleum and natural gas	Mining & Energy	-0.261***	0.043
32	Mining of iron ores	Mining & Energy	-0.244***	0.029
33	Other mining and quarring	Mining & Energy	-0.301***	0.023
34	Electricity production, collection, and distribution	Mining & Energy	-0.345***	0.035
35	Gas production and distribution	Mining & Energy	-0.456***	0.117
36	Processing/preserving of meat	Manufacturing	-0.346***	0.029
37	Processing/preserving of fish	Manufacturing	-0.225***	0.019
38	Processing/preserving of fruit & vegetables	Manufacturing	-0.205***	0.015
39	Vegetable and animal oils and fats	Manufacturing	-0.208***	0.015
40	Dairy products	Manufacturing	-0.280***	0.024
41	Grain mill products	Manufacturing	-0.280***	0.018
42	Starches and starch products	Manufacturing	-0.216***	0.026
43	Prepared animal feeds	Manufacturing	-0.329***	0.024
44	Bakery products	Manufacturing	-0.328***	0.017
45	Sugar	Manufacturing	-0.213***	0.018
46	Cocoa chocolate and sugar confectionery	Manufacturing	-0.316***	0.021
47	Macaroni noodles & similar products	Manufacturing	-0.240***	0.024
48	Other food products n.e.c.	Manufacturing	-0.180***	0.012
49	Distilling rectifying & blending of spirits	Manufacturing	-0.326***	0.030
50	Wines	Manufacturing	-0.379***	0.041
51	Malt liquors and malt	Manufacturing	-0.395***	0.024
52	Soft drinks; mineral waters	Manufacturing	-0.369***	0.021
53	Tobacco products	Manufacturing	-0.246***	0.020
54	Textile fibre preparation; textile weaving	Manufacturing	-0.131***	0.018
55	Made-up textile articles except apparel	Manufacturing	-0.138***	0.028
56	Carpets and rugs	Manufacturing	-0.145***	0.032
57	Cordage rope twine and netting	Manufacturing	-0.181***	0.017
58	Other textiles n.e.c.	Manufacturing	-0.194***	0.024
59	Knitted and crocheted fabrics and articles	Manufacturing	-0.376***	0.030
60	Wearing apparel except fur apparel	Manufacturing	-0.263***	0.028
61	Dressing & dyeing of fur; processing of fur	Manufacturing	-0.499***	0.033
62	Tanning and dressing of leather	Manufacturing	-0.194***	0.029
63	Luggage handbags etc.; saddlery & harness	Manufacturing	-0.431***	0.029
64	Footwear	Manufacturing	-0.146***	0.028
65	Sawmilling and planing of wood	Manufacturing	-0.390***	0.024
66	Veneer sheets plywood particle board etc.	Manufacturing	-0.207***	0.031
67	Builders' carpentry and joinery	Manufacturing	-0.340***	0.025
68	Wooden containers	Manufacturing	-0.379***	0.028
69	Other wood products; articles of cork/straw	Manufacturing	-0.211***	0.020
70	Pulp paper and paperboard	Manufacturing	-0.153***	0.017
71	Corrugated paper and paperboard	Manufacturing	-0.206***	0.020
72	Other articles of paper and paperboard	Manufacturing	-0.061***	0.020

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Table D.3 – continued from previous page

ID	Industry Description	Broad Sector	Estimate	Std. Err.
73	Publishing of books and other publications	Manufacturing	-0.343***	0.029
74	Publishing of newspapers journals etc.	Manufacturing	-0.249***	0.041
75	Publishing of recorded media	Manufacturing	0.321***	0.019
76	Other publishing	Manufacturing	-0.195***	0.030
77	Printing	Manufacturing	-0.337***	0.020
78	Service activities related to printing	Manufacturing	-0.271***	0.022
79	Coke oven products	Manufacturing	-0.244***	0.058
80	Refined petroleum products	Manufacturing	-0.173***	0.013
81	Processing of nuclear fuel	Manufacturing	0.033	0.071
82	Basic chemicals except fertilizers	Manufacturing	-0.110***	0.020
83	Fertilizers and nitrogen compounds	Manufacturing	-0.131***	0.018
84	Plastics in primary forms; synthetic rubber	Manufacturing	-0.230***	0.013
85	Pesticides and other agro-chemical products	Manufacturing	-0.163***	0.018
86	Paints varnishes printing ink and mastics	Manufacturing	-0.243***	0.017
87	Pharmaceuticals medicinal chemicals etc.	Manufacturing	-0.212***	0.015
88	Soap cleaning & cosmetic preparations	Manufacturing	-0.169***	0.013
89	Other chemical products n.e.c.	Manufacturing	-0.179***	0.014
90	Man-made fibres	Manufacturing	-0.066***	0.034
91	Rubber tyres and tubes	Manufacturing	-0.133***	0.020
92	Other rubber products	Manufacturing	-0.253***	0.015
93	Plastic products	Manufacturing	-0.140***	0.013
94	Glass and glass products	Manufacturing	-0.121***	0.014
95	Pottery china and earthenware	Manufacturing	-0.169***	0.019
96	Refractory ceramic products	Manufacturing	-0.290***	0.023
97	Struct.non-refractory clay; ceramic products	Manufacturing	-0.214***	0.025
98	Cement lime and plaster	Manufacturing	-0.222***	0.018
99	Articles of concrete cement and plaster	Manufacturing	-0.341***	0.020
100	Cutting shaping & finishing of stone	Manufacturing	-0.224***	0.023
101	Other non-metallic mineral products n.e.c.	Manufacturing	-0.253***	0.015
102	Basic iron and steel	Manufacturing	-0.245***	0.017
103	Basic precious and non-ferrous metals	Manufacturing	-0.146***	0.025
104	Structural metal products	Manufacturing	-0.232***	0.018
105	Tanks reservoirs and containers of metal	Manufacturing	-0.256***	0.017
106	Steam generators	Manufacturing	-0.241***	0.034
107	Cutlery hand tools and general hardware	Manufacturing	-0.246***	0.020
108	Other fabricated metal products n.e.c.	Manufacturing	-0.205***	0.016
109	Engines & turbines (not for transport equipment)	Manufacturing	-0.253***	0.030
110	Pumps compressors taps and valves	Manufacturing	-0.147***	0.020
111	Bearings gears gearing & driving elements	Manufacturing	-0.244***	0.023
112	Ovens furnaces and furnace burners	Manufacturing	-0.123***	0.037
113	Lifting and handling equipment	Manufacturing	-0.136***	0.017
114	Other general purpose machinery	Manufacturing	-0.230***	0.022
115	Agricultural and forestry machinery	Manufacturing	-0.193***	0.015
116	Machine tools	Manufacturing	-0.195***	0.032
117	Machinery for metallurgy	Manufacturing	-0.253***	0.048
118	Machinery for mining & construction	Manufacturing	-0.101***	0.019
119	Food/beverage/tobacco processing machinery	Manufacturing	-0.288***	0.024
120	Machinery for textile apparel and leather	Manufacturing	-0.328***	0.029
121	Weapons and ammunition	Manufacturing	-0.349***	0.043
122	Other special purpose machinery	Manufacturing	-0.163***	0.029
123	Domestic appliances n.e.c.	Manufacturing	-0.142***	0.017
124	Office accounting and computing machinery	Manufacturing	-0.185***	0.025
125	Electric motors generators and transformers	Manufacturing	-0.207***	0.016
126	Electricity distribution & control apparatus	Manufacturing	-0.143***	0.035
127	Insulated wire and cable	Manufacturing	-0.221***	0.019
128	Accumulators primary cells and batteries	Manufacturing	-0.222***	0.016
129	Lighting equipment and electric lamps	Manufacturing	-0.130***	0.018
130	Other electrical equipment n.e.c.	Manufacturing	-0.192***	0.027
131	Electronic valves tubes etc.	Manufacturing	-0.321***	0.028
132	TV/radio transmitters; line comm. apparatus	Manufacturing	-0.174***	0.025
133	TV and radio receivers and associated goods	Manufacturing	-0.182***	0.024
134	Medical surgical and orthopaedic equipment	Manufacturing	-0.095***	0.024
135	Measuring/testing/navigating appliances etc.	Manufacturing	-0.161***	0.027
136	Optical instruments & photographic equipment	Manufacturing	-0.268***	0.027

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Table D.3 – continued from previous page

ID	Industry Description	Broad Sector	Estimate	Std. Err.
137	Watches and clocks	Manufacturing	-0.438***	0.039
138	Motor vehicles	Manufacturing	-0.281***	0.024
139	Automobile bodies trailers & semi-trailers	Manufacturing	-0.280***	0.023
140	Parts/accessories for automobiles	Manufacturing	-0.207***	0.023
141	Building and repairing of ships	Manufacturing	-0.308***	0.036
142	Building/repairing of pleasure/sport. boats	Manufacturing	-0.356***	0.025
143	Railway/tramway locomotives & rolling stock	Manufacturing	-0.312***	0.023
144	Aircraft and spacecraft	Manufacturing	-0.152***	0.035
145	Motorcycles	Manufacturing	-0.296***	0.030
146	Bicycles and invalid carriages	Manufacturing	-0.205***	0.047
147	Other transport equipment n.e.c.	Manufacturing	-0.382***	0.031
148	Furniture	Manufacturing	-0.194***	0.015
149	Jewellery and related articles	Manufacturing	-0.562***	0.044
150	Musical instruments	Manufacturing	0.293***	0.024
151	Sports goods	Manufacturing	0.246***	0.022
152	Games and toys	Manufacturing	-0.292***	0.044
153	Other manufacturing n.e.c.	Manufacturing	-0.243***	0.018
156	Transport	Services	-0.334***	0.021
157	Travel	Services	-0.221***	0.022
158	Construction	Services	-0.462***	0.033
159	Insurance and pension services	Services	-0.368***	0.027
160	Financial services	Services	-0.301***	0.031
162	Telecommunications, computer, and information services	Services	-0.487***	0.021
163	Other business services	Services	-0.164***	0.025
165	Health services	Services	-0.415***	0.036
166	Education services	Services	-0.253***	0.031

Notes: Estimates are based on author's calculations. This table reports the industry-level estimates (with their corresponding standard errors) of the border effect of the AfCFTA agreement on intra-African trade for each of the 170 industries in the ITPD-E-R02 trade dataset with the full data. These industry-level estimates refer to the coefficients of $BRDR_{AfCFTA_{ij}}$ in Eq. (5) ordered by industry ID (some dropped due to collinearity in border variables caused by missing observations in some industries).

Table D.4: Gravity Estimates of the AfCFTA Border Effects on Intra-African Trade (Excluding Zero Trade Flows)

	(1) Total	(2) Agri	(3) Manu	(4) Mine	(5) Serv
CNTG	0.640 ***	0.566 ***	0.493 ***	0.464 *	0.575 ***
	(0.081)	(0.164)	(0.095)	(0.201)	(0.121)
COLN	0.339 ***	-0.008	0.225 *	0.456	0.613 ***
	(0.098)	(0.186)	(0.114)	(0.237)	(0.152)
DIST	-0.405 ***	-0.756 ***	-0.472 ***	-0.996 ***	-0.318 ***
	(0.046)	(0.073)	(0.054)	(0.092)	(0.053)
LANG	0.257 ***	0.261 *	0.312 ***	0.384 **	0.311 ***
	(0.056)	(0.111)	(0.069)	(0.131)	(0.093)
RTA	0.338 ***	0.885 ***	0.585 ***	0.431 ***	0.241 *
	(0.066)	(0.097)	(0.077)	(0.117)	(0.120)
BRDR_{INTL}	-0.199 ***	-0.193 ***	-0.131 ***	-0.200 ***	-0.258 ***
	(0.007)	(0.011)	(0.008)	(0.012)	(0.010)
BRDR_{AfCFTA}	-0.231 ***	-0.286 ***	-0.179 ***	-0.160 ***	-0.321 ***
	(0.010)	(0.011)	(0.012)	(0.022)	(0.014)
$\text{BRDR}_{AfCFTA \rightarrow ROW}$	-0.485 ***	-0.458 ***	-0.390 ***	-0.376 ***	-0.754 ***
	(0.015)	(0.022)	(0.018)	(0.034)	(0.021)
No of Observations	1160063	299004	599610	195140	66309

Notes: Statistics based on the author's calculations. This table presents the gravity estimates of the AfCFTA Border Effects on intra-African Total, Agriculture (Agri), Manufacturing (Manu), Mining & Energy (Mine), and Service (Serv) trade flows. The estimates are obtained using the PPML estimator with nominal bilateral trade flows (excluding zero trade flows), two-way (exporter-time and importer-time) fixed effects, and clustered standard errors (reported in parentheses) by country pair. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table D.5: Distribution of Border Effect of the AfCFTA at the Industry-level (Excluded Zero Trade Flows)

ID	Industry Description	Broad Sector	Estimate	Std. Err.
1	Wheat	Agriculture	-0.199***	0.024
2	Rice (raw)	Agriculture	-0.202***	0.024
3	Corn	Agriculture	-0.235***	0.022
4	Other cereals	Agriculture	-0.239***	0.023
6	Soybeans	Agriculture	-0.264***	0.042
7	Other oilseeds (excluding peanuts)	Agriculture	-0.299***	0.020
9	Raw and refined sugar and sugar crops	Agriculture	-0.589***	0.122
10	Other sweeteners	Agriculture	-0.274***	0.029
11	Pulses and legumes, dried, preserved	Agriculture	-0.215***	0.021
12	Fresh fruit	Agriculture	-0.315***	0.022
13	Fresh vegetables	Agriculture	-0.353***	0.021
16	Nuts	Agriculture	-0.256***	0.030
19	Eggs	Agriculture	-0.435***	0.041
20	Other meats, livestock products, and live animals	Agriculture	-0.116***	0.027
21	Cocoa and cocoa products	Agriculture	-0.229***	0.045
22	Beverages, n.e.c.	Agriculture	-0.072***	0.033
24	Tobacco leaves and cigarettes	Agriculture	-0.174***	0.020
25	Spices	Agriculture	-0.233***	0.021
26	Other agricultural products, n.e.c.	Agriculture	-0.108***	0.019
29	Mining of hard coal	Mining and Energy	0.010	0.031
31	Extraction of crude petroleum and natural gas	Mining and Energy	-0.206***	0.035
32	Mining of iron ores	Mining and Energy	-0.221***	0.032
33	Other mining and quarrying	Mining and Energy	-0.301***	0.022
34	Electricity production, collection, and distribution	Mining and Energy	-0.331***	0.045
36	Processing and preserving of meat	Manufacturing	-0.300***	0.029
37	Processing and preserving of fish	Manufacturing	-0.187***	0.019
38	Processing and preserving of fruit and vegetables	Manufacturing	-0.205***	0.014
39	Vegetable and animal oils and fats	Manufacturing	-0.202***	0.015
40	Dairy products	Manufacturing	-0.276***	0.023
41	Grain mill products	Manufacturing	-0.264***	0.019
42	Starches and starch products	Manufacturing	-0.207***	0.031
43	Prepared animal feeds	Manufacturing	-0.281***	0.023
44	Bakery products	Manufacturing	-0.318***	0.018
45	Sugar	Manufacturing	-0.199***	0.018
46	Cocoa, chocolate and sugar confectionery	Manufacturing	-0.299***	0.023
47	Macaroni, noodles and similar products	Manufacturing	-0.221***	0.026
48	Other food products, n.e.c.	Manufacturing	-0.177***	0.012
49	Distilling, rectifying and blending of spirits	Manufacturing	-0.255***	0.034
50	Wines	Manufacturing	-0.356***	0.043
51	Malt liquors and malt	Manufacturing	-0.365***	0.025
52	Soft drinks and mineral waters	Manufacturing	-0.349***	0.022
53	Tobacco products	Manufacturing	-0.225***	0.021
54	Textile fibre preparation and weaving	Manufacturing	-0.128***	0.017
55	Made-up textile articles (except apparel)	Manufacturing	-0.137***	0.028
56	Carpets and rugs	Manufacturing	-0.145***	0.032
57	Cordage, rope, twine and netting	Manufacturing	-0.155***	0.017
58	Other textiles, n.e.c.	Manufacturing	-0.172***	0.024
59	Knitted and crocheted fabrics and articles	Manufacturing	-0.344***	0.029
60	Wearing apparel (except fur apparel)	Manufacturing	-0.254***	0.028
61	Dressing and dyeing of fur; processing of fur	Manufacturing	-0.410***	0.033
62	Tanning and dressing of leather	Manufacturing	-0.171***	0.027
63	Luggage, handbags, saddlery and harness	Manufacturing	-0.390***	0.032
64	Footwear	Manufacturing	-0.143***	0.027
65	Sawmilling and planing of wood	Manufacturing	-0.355***	0.026
66	Veneer sheets, plywood, particle board	Manufacturing	-0.181***	0.032
67	Builders' carpentry and joinery	Manufacturing	-0.340***	0.027
68	Wooden containers	Manufacturing	-0.368***	0.029
69	Other wood products; cork and straw articles	Manufacturing	-0.193***	0.020
70	Pulp, paper and paperboard	Manufacturing	-0.147***	0.017
71	Corrugated paper and paperboard	Manufacturing	-0.205***	0.021
72	Other articles of paper and paperboard	Manufacturing	-0.060***	0.020

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Table D.5 – continued from previous page

ID	Industry Description	Broad Sector	Estimate	Std. Err.
73	Publishing of books and other publications	Manufacturing	-0.331***	0.032
74	Publishing of newspapers and journals	Manufacturing	-0.227***	0.038
75	Publishing of recorded media	Manufacturing	0.313***	0.021
76	Other publishing	Manufacturing	-0.175***	0.033
77	Printing	Manufacturing	-0.330***	0.021
78	Service activities related to printing	Manufacturing	-0.243***	0.022
79	Coke oven products	Manufacturing	-0.309***	0.074
80	Refined petroleum products	Manufacturing	-0.169***	0.013
81	Processing of nuclear fuel	Manufacturing	-0.012	0.076
82	Basic chemicals (except fertilizers)	Manufacturing	-0.107***	0.020
83	Fertilizers and nitrogen compounds	Manufacturing	-0.127***	0.016
84	Plastics in primary forms; synthetic rubber	Manufacturing	-0.214***	0.015
85	Pesticides and other agro-chemical products	Manufacturing	-0.143***	0.017
86	Paints, varnishes, printing ink and mastics	Manufacturing	-0.232***	0.017
87	Pharmaceuticals and medicinal chemicals	Manufacturing	-0.202***	0.016
88	Soap, cleaning and cosmetic preparations	Manufacturing	-0.164***	0.013
89	Other chemical products, n.e.c.	Manufacturing	-0.168***	0.013
90	Man-made fibres	Manufacturing	-0.061**	0.024
91	Rubber tyres and tubes	Manufacturing	-0.122***	0.019
92	Other rubber products	Manufacturing	-0.246***	0.016
93	Plastic products	Manufacturing	-0.140***	0.013
94	Glass and glass products	Manufacturing	-0.119***	0.013
95	Pottery, china and earthenware	Manufacturing	-0.158***	0.019
96	Refractory ceramic products	Manufacturing	-0.250***	0.025
97	Structural non-refractory clay products	Manufacturing	-0.202***	0.026
98	Cement, lime and plaster	Manufacturing	-0.230***	0.019
99	Articles of concrete, cement and plaster	Manufacturing	-0.325***	0.021
100	Cutting, shaping and finishing of stone	Manufacturing	-0.198***	0.025
101	Other non-metallic mineral products, n.e.c.	Manufacturing	-0.245***	0.016
102	Basic iron and steel	Manufacturing	-0.242***	0.017
103	Basic precious and non-ferrous metals	Manufacturing	-0.139***	0.025
104	Structural metal products	Manufacturing	-0.223***	0.020
105	Tanks, reservoirs and metal containers	Manufacturing	-0.241***	0.019
106	Steam generators	Manufacturing	-0.178***	0.036
107	Cutlery, hand tools and hardware	Manufacturing	-0.225***	0.022
108	Other fabricated metal products, n.e.c.	Manufacturing	-0.198***	0.016
109	Engines and turbines (non-transport)	Manufacturing	-0.219***	0.036
110	Pumps, compressors, taps and valves	Manufacturing	-0.140***	0.020
111	Bearings, gears and driving elements	Manufacturing	-0.231***	0.023
112	Ovens, furnaces and burners	Manufacturing	-0.085***	0.032
113	Lifting and handling equipment	Manufacturing	-0.118***	0.017
114	Other general purpose machinery	Manufacturing	-0.228***	0.022
115	Agricultural and forestry machinery	Manufacturing	-0.171***	0.014
116	Machine tools	Manufacturing	-0.179***	0.029
117	Machinery for metallurgy	Manufacturing	-0.221***	0.051
118	Machinery for mining and construction	Manufacturing	-0.091***	0.019
119	Food, beverage and tobacco machinery	Manufacturing	-0.266***	0.028
120	Machinery for textile, apparel and leather	Manufacturing	-0.312***	0.031
121	Weapons and ammunition	Manufacturing	-0.283***	0.043
122	Other special purpose machinery	Manufacturing	-0.152***	0.026
123	Domestic appliances, n.e.c.	Manufacturing	-0.142***	0.017
124	Office, accounting and computing machinery	Manufacturing	-0.182***	0.025
125	Electric motors, generators and transformers	Manufacturing	-0.194***	0.015
126	Electricity distribution and control apparatus	Manufacturing	-0.131***	0.035
127	Insulated wire and cable	Manufacturing	-0.202***	0.019
128	Accumulators, primary cells and batteries	Manufacturing	-0.194***	0.016
129	Lighting equipment and electric lamps	Manufacturing	-0.125***	0.018
130	Other electrical equipment, n.e.c.	Manufacturing	-0.186***	0.027
131	Electronic valves and tubes	Manufacturing	-0.295***	0.029
132	TV/radio transmitters and line apparatus	Manufacturing	-0.161***	0.026
133	TV and radio receivers and related goods	Manufacturing	-0.174***	0.024
134	Medical, surgical and orthopaedic equipment	Manufacturing	-0.091***	0.025
135	Measuring, testing and navigating appliances	Manufacturing	-0.155***	0.027
136	Optical and photographic equipment	Manufacturing	-0.233***	0.026

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Table D.5 – continued from previous page

ID	Industry Description	Broad Sector	Estimate	Std. Err.
137	Watches and clocks	Manufacturing	-0.359***	0.044
138	Motor vehicles	Manufacturing	-0.272***	0.024
139	Automobile bodies and trailers	Manufacturing	-0.256***	0.027
140	Parts and accessories for automobiles	Manufacturing	-0.206***	0.023
141	Building and repairing of ships	Manufacturing	-0.220***	0.035
142	Pleasure and sporting boats	Manufacturing	-0.285***	0.025
143	Railway and tramway rolling stock	Manufacturing	-0.261***	0.033
144	Aircraft and spacecraft	Manufacturing	-0.134***	0.036
145	Motorcycles	Manufacturing	-0.236***	0.028
146	Bicycles and invalid carriages	Manufacturing	-0.160***	0.048
147	Other transport equipment, n.e.c.	Manufacturing	-0.328***	0.035
148	Furniture	Manufacturing	-0.192***	0.015
149	Jewellery and related articles	Manufacturing	-0.533***	0.045
150	Musical instruments	Manufacturing	0.302***	0.022
151	Sports goods	Manufacturing	0.250***	0.023
152	Games and toys	Manufacturing	-0.283***	0.047
153	Other manufacturing, n.e.c.	Manufacturing	-0.224***	0.019
156	Transport	Manufacturing	-0.297***	0.017
157	Travel	Services	-0.190***	0.017
158	Construction	Services	-0.427***	0.031
159	Insurance and pension services	Services	-0.314***	0.026
160	Financial services	Services	-0.237***	0.023
162	Telecommunications, computer and information services	Services	-0.450***	0.019
163	Other business services	Services	-0.222***	0.017
165	Health services	Services	-0.422***	0.033
166	Education services	Services	-0.225***	0.023

Notes: Estimates are based on author's calculations. This table reports the industry-level estimates (with their corresponding standard errors) of the border effect of the AfCFTA agreement on intra-African trade for each of the 170 industries in the ITPD-E-R02 trade dataset, excluding zero trade flows. These industry-level estimates refer to the coefficients of $BRDR_{AfCFTA_{ij}}$ in Eq. (5) ordered by industry ID (some dropped due to collinearity in border variables caused by missing observations in some industries).

Table D.6: Welfare Effects of AfCFTA Agreement Across Different Sectors (% Change in Real GDP) with $\sigma = 10$

ISO-3	Country	AGRI (%)	MANU (%)	MINE (%)	SERV (%)
DZA	Algeria	27.59	17.91	10.58	115.35
AGO	Angola	38.14	22.34	3.62	119.09
BEN	Benin	27.07	25.29	88.66	155.81
BWA	Botswana	55.52	33.21	22.20	128.53
BFA	Burkina Faso	32.81	23.15	49.76	56.46
BDI	Burundi	17.85	26.49	52.16	117.21
CMR	Cameroon	22.75	23.14	39.14	140.83
CPV	Cape Verde	27.08	21.25	70.04	109.14
CAF	Central African Republic	45.09	26.16	80.48	117.28
TCD	Chad	45.26	25.32	35.63	132.88
COM	Comoros	11.52	23.07	66.51	111.79
CIV	Côte d'Ivoire	8.64	19.47	23.44	112.99
COD	Democratic Republic of the Congo	46.16	22.64	44.67	121.16
DJI	Djibouti	17.83	16.56	31.54	111.40
EGY	Egypt	8.09	9.63	12.52	19.15
GNQ	Equatorial Guinea	33.90	24.74	16.39	115.83
SWZ	Eswatini	38.84	27.95	52.07	56.39
ETH	Ethiopia	16.65	19.45	55.42	126.27
GAB	Gabon	45.93	24.42	15.07	118.73
GMB	Gambia	32.22	24.22	72.88	86.31
GHA	Ghana	14.31	16.75	15.87	120.99
GIN	Guinea	43.90	21.59	12.30	116.69
GNB	Guinea-Bissau	19.32	23.66	59.52	111.83
KEN	Kenya	13.57	16.98	34.17	26.46
LSO	Lesotho	50.32	35.49	35.31	90.42
LBR	Liberia	29.13	18.23	23.79	122.01
LBY	Libya	41.40	22.81	9.87	124.66
MDG	Madagascar	20.09	23.27	34.85	115.75
MWI	Malawi	24.32	27.93	76.56	117.98
MLI	Mali	19.91	20.72	77.79	121.01
MRT	Mauritania	41.64	22.53	24.81	115.08
MUS	Mauritius	12.97	15.06	14.35	113.92
MAR	Morocco	12.66	9.75	31.78	17.64
MOZ	Mozambique	33.45	28.69	25.58	118.76
NAM	Namibia	38.49	28.37	38.07	64.55
NER	Niger	37.92	25.64	82.79	83.58
NGA	Nigeria	23.08	15.47	4.18	4.16
COG	Republic of the Congo	27.94	22.79	13.89	117.23
RWA	Rwanda	18.65	22.87	29.67	122.73
SEN	Senegal	17.45	16.51	23.95	36.50
SYC	Seychelles	13.96	14.66	27.52	113.14
SLE	Sierra Leone	29.14	24.57	30.29	119.62
SOM	Somalia	36.45	23.65	61.93	124.78
ZAF	South Africa	15.49	7.69	11.86	20.89
SSD	South Sudan	48.35	26.18	24.90	128.04
SDN	Sudan	22.87	23.87	35.45	131.22
STP	São Tomé and Príncipe	32.48	25.06	81.38	117.18
TZA	Tanzania	31.05	22.96	41.49	130.88
TGO	Togo	30.74	24.68	39.51	121.42
TUN	Tunisia	21.11	14.65	44.78	104.93
UGA	Uganda	23.99	22.12	58.49	127.75
ZMB	Zambia	31.31	22.22	43.73	43.83
ZWE	Zimbabwe	26.07	30.62	41.49	129.39
ROW	Rest of World	-0.06	-0.04	-0.00	-0.00

Notes: Estimates are based on author's calculation following the GE PPML procedure prescribed by Anderson et al. (2018). This table shows the welfare effects (percentage change in real GDP) of the AfCFTA agreement corresponding to the elimination of borders between AfCFTA members across different sectors, namely AGRI = Agriculture sector, MANU = Manufacturing sector, MINE = Mining and Energy sector, and SERV = Services sector. All estimates are based on ITPD-E-R02 data with a higher elasticity of substitution of 10. ROW = Rest of World (weighted average for non-AfCFTA countries)

Table D.7: Robustness Checks of Gravity Estimates of the AfCFTA Border Effects on Intra-African Trade

	(1) RTA	(2) + WTO	(3) + PTA	(4) + EIA
CNTG	0.677 ***	0.713 ***	0.726 ***	0.724 ***
	(0.111)	(0.110)	(0.105)	(0.103)
COLN	0.222	0.232 *	0.225 *	0.248 *
	(0.136)	(0.115)	(0.115)	(0.117)
DIST	-0.438 ***	-0.450 ***	-0.454 ***	-0.446 ***
	(0.066)	(0.064)	(0.065)	(0.064)
LANG	0.313 ***	0.232 **	0.232 **	0.247 ***
	(0.074)	(0.072)	(0.072)	(0.071)
RTA	0.972 ***	0.821 ***	0.943 ***	0.674 ***
	(0.087)	(0.086)	(0.171)	(0.161)
WTO		2.779 ***	2.798 ***	2.778 ***
		(0.189)	(0.187)	(0.190)
PTA			-0.147	-0.152
			(0.160)	(0.160)
EIA				0.331 ***
				(0.085)
BRDR_{INTL}	-0.196 ***	-0.187 ***	-0.186 ***	-0.187 ***
	(0.010)	(0.010)	(0.010)	(0.010)
BRDR_{AfCFTA}	-0.244 ***	-0.226 ***	-0.223 ***	-0.213 ***
	(0.014)	(0.014)	(0.014)	(0.014)
$\text{BRDR}_{AfCFTA \rightarrow ROW}$	-0.532 ***	-0.505 ***	-0.501 ***	-0.490 ***
	(0.025)	(0.024)	(0.024)	(0.023)
N	555646	555646	555646	555646

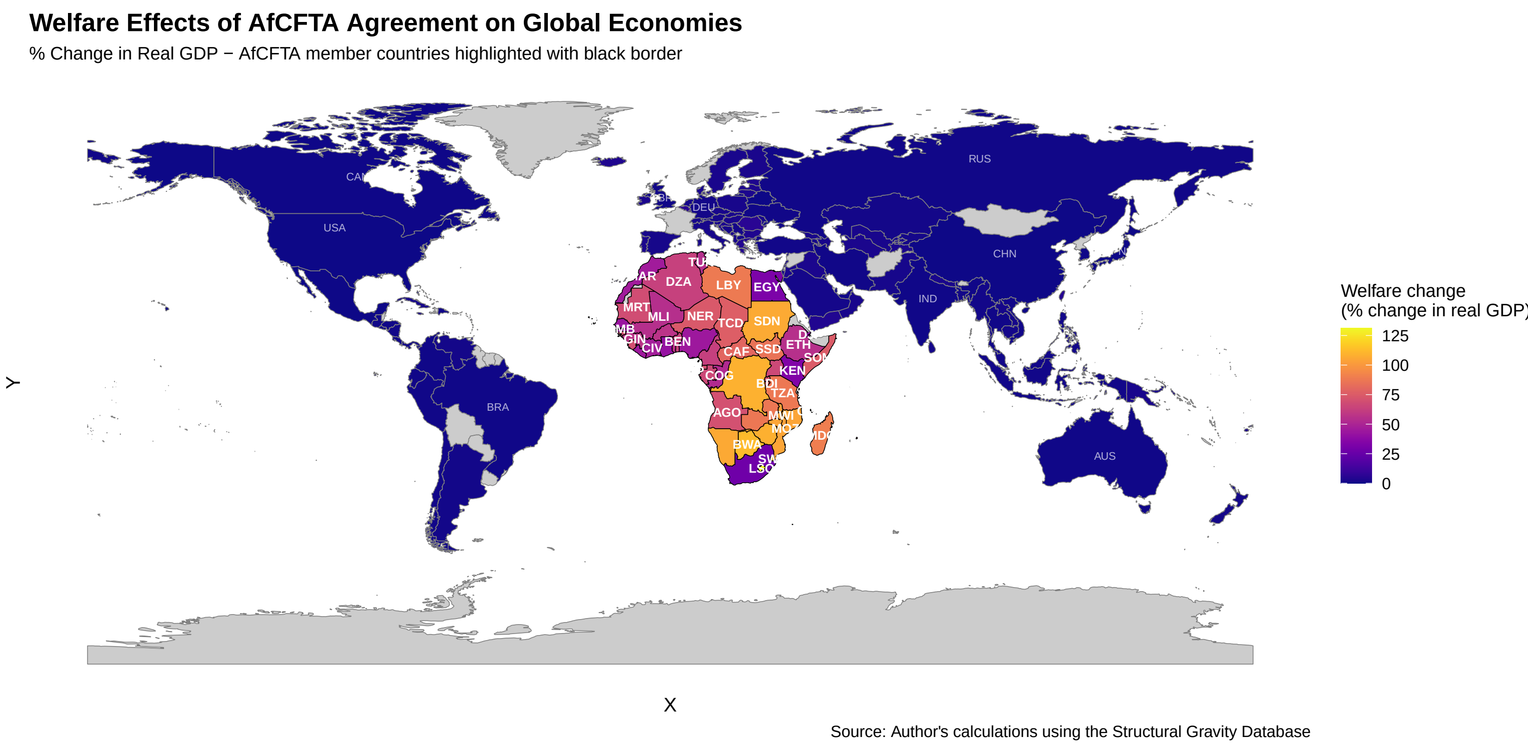
Notes: Statistics based on author's calculations. The table presents the robustness checks of the gravity estimates of ex-ante AfCFTA impacts on intra-African trade under different specifications using the manufacturing trade flows from the Structural Gravity Database. Column (1) is the estimation of the original model with RTA (Regional Trade Agreement) indicator, column (2) adds WTO (World Trade Organization) indicator, column (3) adds WTO (World Trade Organization) and PTA (Preferential Trade Agreement) indicators, and Column (4) adds WTO (World Trade Organization), PTA (Preferential Trade Agreement), and EIA (Economic Integration Agreements) indicators. The estimates are obtained using the PPML estimator with nominal bilateral trade flows, exporter-time and importer-time fixed effects, and clustered standard errors (reported in parentheses) by country pair. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table D.8: Welfare Effects of AfCFTA Agreement Across Different Sectors (% Change in Real GDP) with the Structural Gravity Database

ISO-3	Country	$\sigma = 4$	$\sigma = 5$	$\sigma = 7$	$\sigma = 10$
DZA	Algeria	893.13%	293.97%	62.78%	33.17%
AGO	Angola	1194.91%	368.31%	69.75%	36.50%
BEN	Benin	907.43%	306.57%	63.31%	32.90%
BWA	Botswana	1532.60%	453.71%	111.83%	62.28%
BFA	Burkina Faso	852.62%	291.16%	57.15%	29.11%
BDI	Burundi	1163.58%	380.70%	75.44%	38.10%
CMR	Cameroon	877.20%	296.07%	62.08%	30.73%
CPV	Cape Verde	1044.09%	349.60%	65.10%	30.41%
CAF	Central African Republic	1193.71%	393.19%	79.60%	40.44%
TCD	Chad	1130.21%	375.19%	75.81%	38.42%
COM	Comoros	886.14%	294.47%	53.13%	25.50%
CIV	Côte d'Ivoire	702.45%	242.75%	46.68%	22.19%
COD	Democratic Republic of the Congo	1692.82%	508.65%	108.60%	57.90%
DJI	Djibouti	691.69%	236.17%	39.65%	18.16%
EGY	Egypt	594.66%	201.10%	32.16%	13.68%
GNQ	Equatorial Guinea	1025.93%	342.31%	69.73%	35.11%
SWZ	Eswatini	1502.66%	445.62%	111.53%	62.44%
ETH	Ethiopia	880.95%	294.31%	54.91%	26.67%
GAB	Gabon	999.59%	334.11%	66.38%	33.69%
GMB	Gambia	1030.10%	343.39%	70.43%	35.34%
GHA	Ghana	655.87%	225.18%	40.82%	18.64%
GIN	Guinea	853.35%	290.81%	56.93%	28.69%
GNB	Guinea-Bissau	1070.93%	358.25%	70.28%	34.41%
KEN	Kenya	645.43%	217.29%	33.64%	14.90%
LSO	Lesotho	1751.77%	509.57%	131.00%	72.86%
LBR	Liberia	744.13%	250.84%	44.89%	21.18%
LBY	Libya	1194.97%	384.49%	87.93%	48.48%
MDG	Madagascar	1356.21%	413.63%	89.51%	47.59%
MWI	Malawi	1692.73%	500.81%	102.94%	54.85%
MLI	Mali	838.37%	285.93%	54.62%	27.38%
MRT	Mauritania	1017.21%	340.70%	68.20%	34.27%
MUS	Mauritius	995.57%	312.10%	61.73%	31.94%
MAR	Morocco	691.04%	232.03%	45.58%	21.63%
MOZ	Mozambique	1610.60%	475.88%	104.53%	57.42%
NAM	Namibia	1487.42%	443.60%	105.75%	58.04%
NER	Niger	1056.45%	354.26%	73.64%	38.24%
NGA	Nigeria	753.39%	253.08%	44.18%	20.73%
COG	Republic of the Congo	849.88%	286.04%	52.42%	25.71%
RWA	Rwanda	1098.52%	358.17%	72.27%	36.70%
SEN	Senegal	708.58%	242.84%	42.66%	20.23%
SYC	Seychelles	1020.24%	321.41%	62.19%	31.88%
SLE	Sierra Leone	1026.75%	341.95%	69.17%	34.55%
SOM	Somalia	1128.23%	370.85%	75.24%	37.85%
ZAF	South Africa	680.90%	212.73%	27.71%	10.98%
SSD	South Sudan	1251.51%	407.29%	84.92%	43.37%
SDN	Sudan	1395.78%	443.53%	105.95%	57.66%
STP	São Tomé and Príncipe	1127.94%	372.88%	72.51%	35.72%
TZA	Tanzania	1347.45%	409.97%	87.13%	47.07%
TGO	Togo	848.29%	286.62%	56.99%	29.25%
TUN	Tunisia	806.95%	269.56%	54.52%	28.01%
UGA	Uganda	1003.95%	331.27%	66.12%	33.59%
ZMB	Zambia	1355.55%	407.76%	86.90%	47.43%
ZWE	Zimbabwe	1544.78%	456.64%	108.20%	59.48%
ROW	Rest of World	18.80%	4.03%	0.02%	-0.00%

Notes: Estimates are based on author's calculation following the GE PPML procedure prescribed by Anderson et al. (2018). This table shows the welfare effects (percentage change in real GDP) of the AfCFTA agreement corresponding to the elimination of borders between AfCFTA members using the manufacturing trade flows from the Structural Gravity Database. ROW = Rest of World (weighted average for non-AfCFTA countries).

Figure D.1: Welfare Effects of AfCFTA Agreement on Global Economies



Notes: Figure based on author's calculations. The values in this figure represent the welfare effects of abolishing international borders among African countries resulting from the AfCFTA agreement on manufacturing trade flows based on the Structural Gravity Database with an elasticity of substitution equals 7.